

# Granular Industry Fundamentals and Equity Forecasting

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## Abstract

What can granular industry fundamentals tell us about future equity prices? This paper examines whether real economic information from U.S. industries predicts subsequent equity returns. Using monthly measures of production, employment, hours worked, sales, and inventories across a broad panel of industries, we show that industry-level fundamentals forecast both industry-level equity returns and aggregate market returns, about 4.5 percent, out of sample. The predictive content is strongest in upstream sectors such as manufacturing, transportation, and mining, suggesting that these industries contain early signals about future economic conditions. Different types of fundamentals convey distinct information: production and sales capture demand conditions, while labor-related measures and inventories reflect adjustment dynamics and uncertainty. Overall, the evidence suggests that granular real economic information is incorporated into equity prices only gradually, making industry fundamentals a useful source of information about expected returns.

**Keywords:** industry return predictability; real economic activity; information diffusion; input-output linkages; labor market signals; out-of-sample forecasting; asset pricing

**JEL Classification:** G12, G14, G17, E32, D57

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# Introduction

A firm’s stock price should reflect not only information about the firm itself, but also information about the economic environment in which it operates. Because firms are linked through production networks, news about production, sales, employment, and inventories generated by one part of the economy may affect the value of economically connected firms long before it appears in their own reported fundamentals. Whether financial markets incorporate that information immediately – or only as it gradually propagates through production networks – is an open question with important implications for market efficiency and return predictability. The question is sharpened by the timing of the underlying data: industry-level real-activity releases (production, employment, sales, inventories) arrive far faster than the firm and macro fundamentals used in most return-predictability tests – monthly, with a lag of two days to one month, against a quarter-plus lag elsewhere – so a meaningful share of monthly return variation may be information still in transit rather than noise.

Existing research shows that information diffuses gradually across economically connected firms: customer-supplier relationships, common trading partners, and other one-to-one economic links have all been shown to transmit information into stock prices with delays (Cohen & Frazzini, 2008; Hong et al., 2007; Menzly & Ozbas, 2010). This evidence is largely local, however, examining one type of economic connection at a time. Our contribution is to study diffusion through the entire production network at once: rather than a single link, we trace how real-activity signals originating anywhere in the economy propagate across all 426 industries simultaneously, and ask whether the direction of that propagation reveals the underlying economic mechanism – demand or cost – through which information reaches financial markets. Four literatures connect to one piece of this mechanism but have not previously been combined: production networks (Acemoglu et al., 2012; Gofman et al., 2020; Herskovic et al., 2020; Leontief, 1936), consumption-based asset pricing (Breedon, 1979; Lucas, 1978), gradual information diffusion (Cohen & Frazzini, 2008; Hong et al., 2007; Menzly & Ozbas, 2010), and industry return predictability (Bessembinder et al., 2025; Fama & French, 2012; Hou et al., 2015; Kelly & Pruitt, 2013). The diffusion literature shows prices react slowly but examines single links; the predictability literature finds firm characteristics carry little information, itself suggestive that real activity, not static attributes, is the relevant channel. Neither has been joined to a network structure rich enough to trace diffusion economy-wide.

This paper studies information diffusion at the level of the entire U.S. production network. We construct an Industry Input–Output Propagation Network using 1,049 monthly real-activity signals – production, employment, hours worked, sales, and inventories – across 426 industries (March 1992–October 2024), and evaluate it with Partial Least Squares, both

in sample and out of sample.<sup>1</sup> Unlike traditional input-output analyses, which measure static economic exposure from accounting tables, our network measures how real economic information empirically propagates across industries and whether those propagation patterns predict future returns. The empirical design is fixed *ex ante*, estimated using data only through February 2015, and evaluated exclusively on a completely held-out 2015–2024 period. Relative to prior work on industry predictability and production networks, the contribution is four things taken together: a network built directly from real-activity signals rather than input-output accounting tables alone; a propagation structure whose sign identifies a demand or cost channel rather than an unsigned correlation; an empirical null built by permuting the actual screening procedure rather than a generic significance threshold; and predictability that survives, rather than merely fits, a holdout period never touched during estimation. None of these four is new in isolation; the combination, applied to an economy-wide network rather than a handful of linked firms, is.

The results indicate that production networks contain economically meaningful information about future returns. The network explains 40% of monthly market return variation in sample and retains an out-of-sample  $R^2$  of 23.8% for the aggregate market – large relative to the near-zero benchmark typical of monthly equity return forecasts. Predictability is concentrated in upstream industries such as manufacturing, transportation, and mining and is weak or absent downstream, suggesting that industries positioned earlier in the production process play a larger role in transmitting real economic information to financial markets; the gap between a 41% manufacturing  $R^2$  and a 28% arts-sector  $R^2$  tracks each sector’s distance from final consumers almost exactly.

The evidence also supports an economic interpretation rather than a purely statistical one. The intuition is mechanical rather than exotic: a firm’s suppliers and customers are exposed to its shocks through the goods and services they physically exchange, so real information about one industry’s production, sales, or labor use is relevant to the value of the industries linked to it long before that information shows up in those industries’ own reported fundamentals – markets need only be slow to process the link, not irrational, for this to generate return predictability. Signals associated with production and sales propagate positively, consistent with a demand channel, whereas employment propagates more negatively than hours worked despite the two variables measuring closely related labor inputs, consistent with a cost channel. Because the network is identified by screening 1,049 signals against 20 sectors – 20,980 univariate tests – a natural concern is that some of this predictability re-

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<sup>1</sup>PLS preserves the *sign* of each channel’s propagation as an interpretable linear coefficient, which matters because the paper’s central claim rests on that sign. Table 12 shows alternative linear estimators (Ridge, LASSO, Elastic Net) give materially similar fit, so this choice is not driving the results.

flects statistical selection rather than genuine diffusion. The strongest evidence against that explanation comes from within the data itself: a generic selection procedure has no reason to produce systematically opposite signs for two highly correlated measures of the same labor input, exactly as the demand-versus-cost distinction predicts in advance. A permutation test that reconstructs the entire screening procedure under an empirical null corroborates this directly: the observed concentration of predictive power in broadly-propagating signals is essentially absent by chance (Section 6.2.1); Section 6.2.2 discusses further robustness dimensions.

Section 1 describes the data. Section 2 presents the empirical methodology. Section 3 presents the main results. Section 4 concludes.

# 1 Data

This study uses data from four main sources. First, industry returns (Section 1.1), constructed from CRSP firm-level data and aggregated to NAICS sectors. Second, extensive and intensive human capital measures from the Current Employment Statistics (CES) program (Section 1.2), covering employment levels and average weekly hours worked across 426 industries. Third, industrial production indices from the Federal Reserve (Section 1.3), measuring physical output across 61 manufacturing, mining, and utilities sectors. Fourth, sales and inventory data from three supply chain stages (Section 1.4): manufacturing (from the Census Bureau’s M3 Survey), wholesale trade (from the Wholesale Trade Survey), and retail trade (from the Monthly Retail Trade Survey). Together, these sources provide 1,049 fundamental signals that form the input-output predictability network. The following subsections detail each data source, their construction, transformations, and coverage.

These four sources were chosen over other commonly used predictors – prices, earnings, purchasing managers’ indices (PMI), or credit spreads – for three reasons specific to this paper’s question. First, they measure *real* activity directly: employment, hours, physical production, sales, and inventories are quantities, not prices, so using them avoids the circularity of predicting one financial-market variable (returns) with another (e.g., credit spreads, which are themselves asset prices reflecting the market’s own risk assessment). Second, they are collected independently of financial markets, by government statistical agencies for administrative and survey purposes, so any predictive content they carry cannot be mechanically inherited from prior price movements. Third, they are available at the granularity this paper’s network requires: 426 industries for employment and hours, and 61 to 77 industries for the remaining sources, versus PMI, which is published as a small number of national or broad-sector composite indices unsuited to constructing an industry-by-industry

network. Earnings and analyst forecasts were considered and excluded for a related reason: both are already filtered through firms’ own disclosure choices and analysts’ interpretation, introducing a layer of pricing-relevant processing that this paper’s real-activity signals are constructed specifically to avoid.

## 1.1 Industry Returns

Industry returns are constructed from CRSP monthly firm-level data, aggregated to the 19 main NAICS sectors plus a value-weighted aggregate market portfolio, following standard Fama-French construction. NAICS is used in preference to alternative classifications (SIC, GICS, Fama-French industry groupings, or text-based classifications) because it groups firms by production-process similarity, which is the natural basis for an input-output propagation network; Appendix 6.1 discusses this choice and the underlying classification systems in detail, including how the paper’s 19-sector-plus-market convention relates to the 20 sectors defined by the official NAICS classification. The sample covers March 1992 to October 2024 (392 monthly observations). Table 3 summarizes each sector’s risk-return moments and systematic risk; sector market capitalization is highly concentrated (Information and Manufacturing alone account for roughly half of total market value), and betas range from 0.49 (Utilities) to 1.39 (Construction). Table 4 reports the corresponding correlation structure, and the underlying BEA input-output linkages used to construct the network are summarized in Appendix 6.1.

## 1.2 Human Capital

Industry-level human capital data come from the BLS Current Employment Statistics (CES) program, a monthly survey of approximately 121,000 businesses covering roughly one-third of nonfarm payroll jobs, released on the first Friday of the following month. CES provides two distinct margins for 426 industries: *extensive margin* (employment levels), reflecting hiring and firing decisions that signal longer-term commitments, and *intensive margin* (average weekly hours), reflecting more flexible near-term production adjustments (Hamermesh, 1993). As shown in Section 3, this distinction manifests in differential propagation patterns: extensive human capital propagates more negatively than intensive human capital, consistent with a cost-channel interpretation. Employment levels are transformed into log differences and hours into first differences, both winsorized at the 1st and 99th percentiles, with a 1-month lag reflecting the timely release of CES data. Full construction detail is in Appendix 6.1.

### 1.3 Industrial Production

Industrial Production (IP) data from the Federal Reserve (FRED) comprise 61 seasonally adjusted series spanning manufacturing, mining, and utilities, reported as an index (2017=100) and covering January 1972 to August 2025. Raw index levels are transformed into log differences, winsorized at the 1st and 99th percentiles, and lagged two months to account for the data release schedule. Full series composition and summary statistics are in Appendix 6.1.

### 1.4 Sales and Inventories

Sales and inventory data are drawn from three stages of the supply chain: manufacturing shipments and inventories (Census M3 Survey, 40 variables), wholesale trade sales and inventories (Census Wholesale Trade Survey, 42 variables), and retail trade sales, inventories, and inventory-to-sales ratios (Census Monthly Retail Trade Survey, 54 variables). All three sources are reported as percentage changes (manufacturing and retail) or transformed into log differences (wholesale), winsorized at the 1st and 99th percentiles, and lagged two months. Full construction detail, industry coverage, and summary statistics for each source are in Appendix 6.1.

### 1.5 Data Overview

Table 7 provides a comprehensive overview of all datasets used in this study. The return matrix, which serves as the prediction target, has dimensions  $392 \times 20$ , representing 392 monthly observations across 20 industry sectors from March 1992 to October 2024. All macroeconomic and fundamental variables are consolidated into a single input-output matrix with dimensions  $392 \times 1,049$ , where each column represents a time series signal. This matrix comprises 426 employment series, 426 hours worked series, 61 industrial production indices, 40 manufacturing variables (shipments and inventories), 42 wholesale trade variables (sales and inventories), and 54 retail trade variables (sales, inventories, and inventory-to-sales ratios). Each signal in this matrix serves as a potential predictor of industry returns, forming the basis for the univariate and multivariate predictability analyses that follow.

### 1.6 Factor Intensity Analysis

Table 8 presents the factor intensities of industries, revealing substantial cross-sectoral heterogeneity. This heterogeneity in production functions creates differential sensitivities to shocks in labor markets, capital availability, materials costs, and energy prices, which should manifest in heterogeneous return predictability. Accommodation is the sector with highest

labor intensity (0.91), substantially above the mean of 0.66, with below-average capital intensity (1.2 vs. 2.39) but above-average material intensity (0.65 vs. 0.47) and around-average energy intensity (0.35 vs. 0.33). At the other extreme, utilities exhibits the lowest labor intensity (0.33) but the highest capital intensity (6.5) and energy intensity (0.95). Mining and transportation also show high capital and energy intensities, reflecting their reliance on heavy machinery and fuel consumption. At the opposite end, professional services shows the lowest capital intensity (0.2) along with low material and energy intensities, consistent with knowledge-based production. Retail trade shows high labor intensity (0.83) combined with the highest material intensity (0.75), reflecting the labor-intensive nature of distributing goods. Arts, entertainment, and recreation displays high capital intensity (3.5) relative to other services, likely reflecting investment in venues and equipment.

## 2 Empirical Framework

The data collected above form an input-output matrix with dimensions  $T \times S = 392 \times 1,049$ , representing 392 monthly observations across 1,049 fundamental signals, along with 20 sets of time series (19 NAICS sectors plus the aggregate market portfolio) whose networks are identified within this matrix. This section presents the empirical methodology for estimating input-output propagation networks and modeling expected sector returns as a function of their networks' input-output fundamentals.

### 2.1 Notation and Setup

**Dependent Variables.** The object of interest is industry excess returns:

- Let  $t \in \{1, \dots, T\}$  index monthly periods, where  $T = 392$  spanning March 1992 to October 2024.
- Let  $j \in \{1, \dots, J\}$  index sectors, where  $J = 20$  in our analysis: 19 NAICS sectors and the aggregate market portfolio.
- $r_{j,t}$ : Excess return of sector  $j$  in month  $t$  over the risk-free rate
- $\mathbf{r}_t = (r_{1,t}, \dots, r_{J,t})'$ :  $J$ -dimensional vector of sector returns at time  $t$
- $\mathbf{R} \in \mathbb{R}^{T \times J}$ : Return matrix containing the full panel of sector returns

**Independent Variables.** The predictor variables consist of fundamentals organized into an input-output matrix:

- Let  $s \in \{1, \dots, S\}$  index the fundamental signals in the input-output matrix, where  $S = 1,049$
- $x_{s,t}$ : Fundamental signal  $s$  at time  $t$ , lagged appropriately relative to returns (1-month lag for CES variables, 2-month lag for all others)
- $\mathbf{x}_t = (x_{1,t}, \dots, x_{S,t})'$ :  $S$ -dimensional vector of all signals at time  $t$
- $\mathbf{X} \in \mathbb{R}^{T \times S}$ : Input-output (predictor) matrix containing the full panel of fundamental variables

## 2.2 Network Structure and Identification

**Network Definition.** Each signal in the input-output matrix  $\mathbf{X}$  is assumed to contain information about next period returns of a set of sectors, denoted as  $\mathcal{J}^s \subseteq \{1, \dots, J\}$ . The cardinality  $|\mathcal{J}^s|$  represents signal  $s$ 's *propagation score*, measuring how many sectors signal  $s$  predicts. Signals with larger propagation scores predict returns across many sectors and therefore capture more systematic risk factors.

Conversely, for each sector  $j$ , its *propagation network* is defined as the set of signals that predict its returns, denoted as  $\mathcal{S}^j \subseteq \{1, \dots, S\}$ . The cardinality  $|\mathcal{S}^j|$  measures the number of signals in sector  $j$ 's network. Sectors that provide goods or services to a variety of industries, or purchase inputs from a variety of industries, are expected to have higher  $|\mathcal{S}^j|$ —their returns are sensitive to a broader set of economic fundamentals.

These definitions establish a duality:  $s \in \mathcal{S}^j$  if and only if  $j \in \mathcal{J}^s$ . This dual perspective allows us to analyze networks from both the perspective of receiving sectors (which signals matter for each sector?) and transmitting signals (which sectors does each signal affect?).

**Univariate Signal Screening.** To identify the network structure, univariate time series regressions are used first to test whether each signal  $s$  predicts each sector  $j$ 's returns. For each sector-signal pair  $(j, s)$ , the following is estimated:

$$r_{j,t+1} = \theta_j + \gamma_{j,s} x_{s,t} + \nu_{j,t+1} \quad (1)$$

where  $\theta_j$  is a sector-specific intercept,  $\gamma_{j,s}$  is the propagation coefficient measuring the information about next period return of sector  $j$  revealed by signal  $s$ , and  $\nu_{j,t+1}$  is an error term. The magnitude of  $\gamma_{j,s}$  can be zero, positive, or negative.

Signals are classified based on their propagation patterns:

- **Noise signals:** If  $|\mathcal{J}^s| = 0$  (i.e.,  $\gamma_{j,s} = 0$  for all  $j$ ), signal  $s$  contains no predictive information.
- **Systematic signals:** If  $\gamma_{j,s} \neq 0$  for the market portfolio ( $j = \text{market}$ ), signal  $s$  is called a *systematic signal*. These signals predict aggregate market returns and therefore capture economy-wide risk factors.
- **Sector-specific signals:** If  $\gamma_{j,s} \neq 0$  for some sector  $j \neq \text{market}$  but  $\gamma_{\text{market},s} = 0$ , signal  $s$  is called a *sector-specific signal*. Note that a signal can predict several sectors (have a high propagation score) but still be sector-specific if it does not predict the market portfolio.

The network  $\mathcal{S}^j$  for sector  $j$  is then defined as the set of signals for which  $|\text{t-stat}(\hat{\gamma}_{j,s})| > c$ , where  $c$  is a significance threshold (e.g.,  $c = 1.96$  for 5% significance). Similarly,  $\mathcal{J}^s$  is the set of sectors for which signal  $s$  has significant predictive power.

**Multiple Testing.** Constructing the network requires  $J \times S = 20 \times 1,049 = 20,980$  univariate hypothesis tests. Applying a nominal 5% significance threshold uniformly across this many tests is known to substantially overstate the number of genuine relationships, since approximately  $0.05 \times 20,980 \approx 1,049$  significant results would be expected by chance alone even if no predictor contained any information (Harvey et al., 2016). Section 6.2.1 addresses this concern with a bootstrap procedure that constructs the empirical null distribution of every reported statistic—the aggregate count of significant links, the industry-level share of significant predictors, and each predictor’s propagation score—directly from the data, by independently permuting the time ordering of each industry’s returns while holding the predictor panel fixed. This bootstrap is, in effect, a resampling-based multiple-testing correction: rather than relying on the nominal 5% threshold, each observed statistic is compared to its exact empirical distribution under the null of no predictability, which automatically accounts for the cross-sectional dependence among the 1,049 predictors and 20 industry returns.

**Network Interpretation.** The network structure need not be fully aligned with traditional input-output linkages. Market values are set by investors who may be subject to behavioral biases—a signal from an industry that is not in the input-output linkages of sector  $j$  may still contain information about the market value of  $j$  due to overreaction, misinterpretation, or other market frictions. Conversely, a signal from an industry that is in the

input-output linkages of sector  $j$  may not predict its returns if the relationship is already fully incorporated into prices. The propagation of input-output fundamentals to market values of sectors is therefore determined empirically rather than imposed a priori.

## 2.3 Multivariate Model Specification

**Static Network Assumption.** For simplicity, the network structure and propagation coefficients are assumed to be static—that is, the strength of signals does not change over time. This assumption allows for consistent estimation of network linkages and provides a clear interpretation of long-run propagation patterns. While recent work on dynamic networks (Gofman et al., 2020) suggests that interconnections may evolve over time, a static network provides a natural benchmark and is consistent with the relatively stable nature of industrial organization and input-output relationships over the business cycle. Future work could relax this assumption by allowing time-varying networks  $\mathcal{S}_t^j$  estimated using rolling windows or state-space methods.

**Linear Model.** Under the static network assumption, the realized return for sector  $j$  is given by:

$$r_{j,t+1} = \alpha_j + \sum_{s \in \mathcal{S}^j} \beta_{j,s} x_{s,t} + \varepsilon_{j,t+1} \quad (2)$$

where  $\alpha_j$  is a sector-specific intercept capturing the unconditional expected return,  $\beta_{j,s}$  measures the sensitivity of sector  $j$ 's returns to signal  $s$  (which may differ from the univariate coefficient  $\gamma_{j,s}$  due to multivariate interactions), and  $\varepsilon_{j,t+1}$  is an unpredictable shock satisfying  $E[\varepsilon_{j,t+1} \mid \mathcal{F}_t] = 0$ , with  $\mathcal{F}_t$  denoting the information set available at time  $t$ .

Let  $\mathbf{X}^j \in \mathbb{R}^{T \times |\mathcal{S}^j|}$  denote the submatrix of  $\mathbf{X}$  containing only the signals in sector  $j$ 's network. The model can be written in matrix form as:

$$\mathbf{r}^j = \alpha_j \mathbf{1}_T + \mathbf{X}^j \boldsymbol{\beta}_j + \boldsymbol{\varepsilon}^j \quad (3)$$

where  $\mathbf{r}^j = (r_{j,2}, \dots, r_{j,T+1})'$  is the vector of returns for sector  $j$ ,  $\mathbf{1}_T$  is a  $T$ -dimensional vector of ones,  $\boldsymbol{\beta}_j = (\beta_{j,s_1}, \dots, \beta_{j,s_{|\mathcal{S}^j|}})'$  is the vector of coefficients for sector  $j$ 's network signals, and  $\boldsymbol{\varepsilon}^j$  is the error vector.

**Economic Interpretation.** The coefficient  $\beta_{j,s}$  quantifies how a one-standard-deviation change in fundamental  $s$  translates into expected returns for sector  $j$ . A positive coefficient indicates that increases in the fundamental predict higher future returns, while a negative

coefficient suggests the opposite. The sign and magnitude of  $\beta_{j,s}$  reflect the underlying economic mechanism through which signal  $s$  affects sector  $j$ . For instance, positive coefficients on own-sector employment may reflect demand growth, while negative coefficients on upstream inventories may signal supply constraints.

## 2.4 Theoretical Predictions

Based on economic theory and the propagation channels discussed in the introduction, the following hypotheses are derived about the signs of propagation coefficients:

**Production and Sales Signals.** Industrial production and sales are both signals of aggregate demand for sector  $j$ 's output (Acemoglu et al., 2012; Keynes, 1936; Leontief, 1936). They directly affect the sector's revenue and therefore should have positive propagation coefficients. For example, an observed increase in production of manufacturing of heavy equipment is a positive signal on future revenue and therefore market value of the transportation sector—as transportation supplies to manufacturing of equipment.

**Inventory Signals.** Inventories signal negatively because they show a deficit in aggregate demand and therefore a decrease in the market value of the sector (Khan & Thomas, 2007). When inventories accumulate, it indicates that production exceeds final demand, signaling weak demand that will reduce future production and input purchases.

**Human Capital Signals.** The propagation coefficient of intensive and extensive human capital could be positive or negative depending on the functionality of labor in the industry from which signal  $s$  originates (Blanchard, 1986; Blinder, 1982). If labor is used primarily for maintenance, capital upkeep, and other forms of cost (as in utilities or mining), increases in employment or hours propagate negatively to connected sectors through the cost channel. On the other hand, if labor is directly involved in the production of goods or services (as in manufacturing or retail), increases in employment or hours signal rising demand and therefore contain positive information about the market value of connected sectors through the demand channel.

Extensive human capital (employment levels) should also propagate more negatively than intensive human capital (hours worked), since hiring new employees entails higher fixed costs and signals longer-term commitments, while asking existing employees to work longer hours is more flexible and reflects near-term production needs (Hamermesh, 1993).

## 2.5 Estimation Methods

Given the high dimensionality of the network ( $|\mathcal{S}^j|$  can be large, often 50–100 signals per sector), standard OLS estimation may suffer from multicollinearity and overfitting, so several estimation methods are employed:

**Ordinary Least Squares (OLS).** For sectors with relatively small networks or when focusing on a subset of signals, equation (2) is estimated using OLS with Newey-West HAC standard errors (6 lags) to account for heteroskedasticity and autocorrelation.

**Regularization Methods.** For sectors with large networks, regularization techniques are employed:

- **Ridge Regression:** Adds an L2 penalty to shrink coefficients toward zero, reducing multicollinearity while retaining all signals.
- **LASSO:** Adds an L1 penalty that sets some coefficients exactly to zero, performing automatic variable selection.
- **Elastic Net:** Combines L1 and L2 penalties, balancing between Ridge and LASSO.

**Partial Least Squares (PLS).** PLS reduces the dimensionality of the predictor space by constructing linear combinations of signals that maximize covariance with returns, rather than only explaining predictor variance (as PCA does) or shrinking individual coefficients toward zero (as LASSO and Elastic Net do). This distinction is economically, not just statistically, motivated. Principal components analysis is unsupervised: it finds the directions of greatest variation in the 1,049 signals themselves, which need not align with the directions that predict returns, and offers no reason to expect its leading components to correspond to the demand and cost channels this paper’s theory predicts. LASSO and Elastic Net operate signal-by-signal, zeroing out or shrinking individual coefficients; they are well suited to variable selection (and are used for exactly that purpose in Table 15 and throughout the paper) but do not, by construction, produce a small number of interpretable composite factors. PLS does: because each component is a covariance-maximizing linear combination of the underlying signals, it is a natural fit for a paper whose central claim is that the demand and cost channels operate as coherent, sign-consistent factors across many correlated signals, rather than as a small set of individually important variables. Table 12 confirms this choice is not idiosyncratic to the results: Ridge, LASSO, and Elastic Net applied to the same networks produce materially similar in-sample fit to PLS, so the headline findings do

not depend on using PLS specifically rather than another linear method – PLS is preferred for interpretability of the channel structure, not because it is the only estimator that fits the data.

The number of PLS components is fixed *ex ante* at 2 for every out-of-sample specification in this paper, rather than chosen by cross-validation or tuned on the test period. With 274 training months and up to 65 predictors, each additional component consumes scarce time-series degrees of freedom, so a small, fixed number is a conservative default rather than a tuned maximum. Prior applications of PLS to monthly return prediction with similarly high-dimensional predictor sets, such as Kelly and Pruitt (2013), likewise retain only a small number of components (one to a few), which this choice follows. The main reason for fixing the number in advance, rather than selecting it to maximize in-sample or test-period fit, is that doing so is the safeguard against data snooping described in the Introduction. Specification D in Table ?? shows what happens when this discipline is relaxed and components are tuned on the test window instead; the resulting number is explicitly flagged as contaminated rather than reported as a robustness check. A full sensitivity table varying the component count from 1 to 5 under the same *ex-ante* discipline is a natural extension, included among the specified checks in Section 6.2.2.

The economic interpretation of the two retained components is, at this stage, a hypothesis rather than a verified decomposition: given the theory in Section 1.2 and the sign pattern documented in Section 3, the natural conjecture is that the first component loads primarily on the broad demand-channel signals (production, sales) and the second contrasts against the cost-channel signals (employment, hours), but confirming this requires inspecting the actual component loadings rather than only the fitted  $R^2$ , which Table 20 lists as a specified extension.

### 3 Results

The previous sections assembled the input–output matrix and the framework for constructing the **Input–Output Predictability Network**. This involves (i) running univariate regressions to identify predictive links across industries and (ii) modeling expected returns as a linear function of the networks’ associated inputs and outputs. This section reports results in that order: first how inputs and outputs propagate to market values across industries, then the predictability of the resulting linear model.

### 3.1 Input–Output Propagation Network

This subsection begins with a big-picture view of the signals and their predictive power. Panel A of Table 9 summarizes these results: 1,049 predictors send signals to 20 sectors, creating 20,980 possible input–output connections. Of these, 1,360 (6.5%) are statistically significant, while the rest are classified as noise. The distribution of signs is nearly balanced—52.3% positive and 47.7% negative—suggesting that input–output propagation may operate as a *zero-sum mechanism* in the short run, where gains in some sectors are offset by losses in others.

Panel B breaks down accuracy and sign patterns across types of inputs and outputs. **Extensive human capital** produces 6.5% accurate signals, with 43% positive and 57% negative, consistent with theories emphasizing the dual role of employment in affecting both costs and demand. **Intensive human capital** shows slightly lower accuracy (5.9%) but a more positive bias (55% positive, 45% negative), consistent with cost-based interpretations—since increasing hours for existing employees is typically less costly than hiring new ones.

**Industrial production** data include 61 goods, yielding 1,220 sector-level regressions, of which 92 (7.5%) are significant. About 75% of these signals are positive, consistent with the view that production reflects aggregate demand and thus predicts market values positively.

**Sales** of manufacturers, wholesalers, and retailers (manufacturing shipments, wholesale sales, and retail sales; 77 signals) show 1,540 estimations, with 110 significant—an accuracy rate of roughly 7.1%. About 94% of these signals are positive, consistent with sales being a strong indicator of demand that propagates primarily in a positive direction.

Finally, **inventories** (manufacturing, wholesale, and retail inventories plus the retail inventory-to-sales ratio; 59 signals) show 1,180 estimations, with 98 significant, an accuracy rate of roughly 8.3%, with roughly 80% negative coefficients, in line with theories that inventories signal demand–supply imbalances and thus predict sectoral market values negatively.

#### 3.1.1 Origins of Propagation $|\mathcal{J}^s|$

Beyond aggregate statistics, the distribution of signal score—defined as the number of sectors each signal predicts—reveals important structural features of the input–output propagation network. Figure 1 presents this distribution. Of the 1,049 predictors, 53.1% (557 signals) fail to significantly predict any sector’s returns and are identified as noise. Among the remaining 46.9% (492 signals) with at least one significant relationship, the distribution is right-skewed: 20.4% (214 signals) predict exactly one sector (purely idiosyncratic signals), and another 26.5% (278 signals) predict two or more. Predictive power turns out to be concentrated, but in a specific way rather than in a handful of dominant signals. Only 24 predictors

(2.3% of the full panel) predict at least 10 of the 20 sectors, with the single strongest signal predicting 15. The network’s systematic backbone therefore rests on a small, identifiable core of broadly-propagating signals layered on top of a long tail of narrow, sector-specific ones. (Table 19 in Appendix 6.2.1 reports the complete score-by-score breakdown underlying these figures, together with the corresponding permutation-based significance test.)

Panel B of Figure 1 illustrates the directional consistency of signal propagation. Among the 492 signals with non-zero predictive power, the vast majority (97.0%) exhibit unidirectional propagation: 266 signals (54.1%) send exclusively positive signals, while 211 (42.9%) send exclusively negative ones. Only fifteen signals (3.0%) display mixed propagation patterns, predicting positive returns for some sectors and negative returns for others. This directional consistency suggests that each fundamental signal carries a coherent economic interpretation—either as “good news” or “bad news”—that propagates uniformly through the network.

Table 10 presents the 40 strongest signals – 20 with the highest positive scores and 20 with the highest negative scores – and the pattern across them is economically coherent rather than idiosyncratic. Positive signals cluster around consumer and capital-investment demand: production and employment in food manufacturing, durable goods, and specialized equipment. Negative signals cluster around cost pressure and demand deficiency: warehousing and inventories, and labor-intensive service categories such as home health care and caregiving. One negative signal is worth singling out. Hours worked in waste collection predict lower returns network-wide, echoing Savov (2011), who use garbage production as a consumption proxy and find higher consumption associated with lower expected returns through investors’ marginal utility; this paper’s result offers a novel, monthly, tradeable version of that same proxy.

### 3.1.2 Destination of Propagation $J^S$

Table 11 shows the other side of the network: how many signals predict each sector. This ranges from 57 (Other Services, 5.4% of the 1,049 signals) to 85 (Arts, 8.1%), with each signal explaining 1.27% of monthly variation on average.

The sign of these predictors is also sector-specific. Others (Berkshire Hathaway), Utilities, Wholesale, Agriculture, and Transportation are each over 70% positively predicted – a pattern consistent with their upstream or intermediating position in the supply chain, where expansion signals from elsewhere in the economy raise their own expected returns. Arts, Retail, and Construction are each over 70% negatively predicted, consistent with their downstream, consumer-facing position, where the same expansion signals more often reflect rising costs passed down to them. The market portfolio itself is almost exactly balanced

(50.8% positive, 49.2% negative), as expected for a value-weighted aggregate of both patterns.

### 3.1.3 Propagation Heatmaps

Figures visualizing the full propagation structure for each data source (which sectors predict which, and with what sign) are collected in Appendix 5.3.

## 3.2 Multivariate Regression Results

The univariate analysis identified an average of 68 network signals per sector, each explaining 1.27% of monthly return variation on its own; multicollinearity among them raises the question of whether they carry independent information jointly. Table 12 reports in-sample  $R^2$  from four estimation methods (PLS, Ridge, LASSO, Elastic Net) applied to each sector’s network on the full 1992–2024 sample. All four give materially similar fit, so the discussion below uses PLS (2 components) for predictability and LASSO for variable retention. A linear model of each sector’s returns on its own network explains 34% of monthly variation on average across sectors, and 40% for the aggregate market – both economically large for monthly equity variation and statistically credible given 68 signals is not high-dimensional relative to the sample.

Predictability is highest for Manufacturing (41%), the market (40%), Information (41.2%), and Utilities (39.7%), sectors that are either large, data-rich, or structurally central as suppliers to many other industries. It is lowest for Arts, Retail, Construction, and Accommodation, which are predominantly downstream, consumer-facing sectors. LASSO retains 51 of the average 68 network predictors (75%), removing the rest as redundant; this parsimony, combined with a mean retained-model  $R^2$  of 36.3%, indicates that most of the identified network genuinely carries independent information rather than being an artifact of correlated near-duplicates. Manufacturing’s own network illustrates this starkly: LASSO zeros out only 10 of 67 signals, and the market’s 65-signal systematic network loses only 10 as well. This suggests that sector-specific signals contribute comparatively little beyond whatever independent information the systematic signals already contain, a hypothesis tested directly in the next subsection.

### 3.2.1 Systematic vs. Sector Specific Signals

Table 13 elaborates on the structure of the network across sectors, decomposing the total number of signals into *systematic* signals (reported in the second column) and *sector-specific*

signals (reported in the last major column), with subcolumns indicating how many of these signals are shared with other sectors.

The Utilities sector has 81 signals, of which only 9 are systematic—shared with the aggregate market. The remaining 72 signals are predominantly sector-specific: 39 are unique to Utilities, 10 are shared with one other sector, 8 with two sectors, 5 with three sectors, and 10 with more than three sectors. This composition makes Utilities one of the most distinctive sectors in terms of network structure. Education, Arts, Agriculture, Others, and Mining are other sectors characterized by a large share of sector-specific signals.

In contrast, sectors such as Manufacturing, Finance, Professional Services, and Real Estate exhibit the fewest sector-specific signals, suggesting that their return dynamics are more tightly linked to systematic, economy-wide forces.

Table 14 reports predictability results separately for systematic and sector-specific components. The first three columns show the total number of signals, as well as the breakdown into systematic and sector-specific signals. The next three columns present their corresponding predictive performance, estimated using Partial Least Squares (PLS).

Several notable results emerge. First, systematic and idiosyncratic (sector-specific) signals each explain, on average, 21–23% of the time-series variation in sector returns. When combined, they explain approximately 34%, with the remaining variation attributable to the joint component shared between the two sets.

Second, there is substantial heterogeneity across sectors in terms of the dominant source of predictability. Some sectors are primarily driven by systematic signals, while others are better explained by sector-specific ones. Among the systematically driven sectors are Manufacturing—where nearly all predictability (37% out of 41%) comes from systematic signals—as well as Information, Finance, Administration, Transportation, Real Estate, and Others.

Conversely, Utilities is the sector whose predictability is almost entirely driven by sector-specific signals. Following Utilities, sectors such as Education, Agriculture, and Health also exhibit dominance of sector-specific information in explaining their return dynamics.

### 3.3 Reconciling Out-of-Sample Predictability for the Aggregate Market

The in-sample results above use the full sample (1992–2024) and are therefore subject to the usual concern that predictability estimated and evaluated on the same data overstates genuine forecastability. The specifications reported later in this paper vary along three dimensions – the number and selection of predictors, the estimator (OLS versus PLS), and

the out-of-sample evaluation protocol – so this subsection assembles every out-of-sample (OOS) estimate for the aggregate market portfolio reported elsewhere in the paper into a single table, states what each specification holds fixed, and designates one specification as the paper’s primary out-of-sample claim.

The headline out-of-sample result is Specification C: all 47 systematic signals under PLS, with the number of components fixed *ex ante* at 2 (avoiding any use of the test window in model selection), evaluated once on a 70/30 chronological split (train: May 1992–Feb 2015; test: Mar 2015–Dec 2024). It reaches 23.8% out-of-sample  $R^2$  for the aggregate market. This is the number reported in the abstract and used throughout the rest of the paper; Table ?? reports three further specifications for robustness, not as competing headlines. The most direct answer to whether the result survives a substantially smaller predictor set is yes, though attenuated: a smaller 8-signal Lasso specification under OLS still reaches 9.1% out-of-sample  $R^2$ , and a 20-signal specification reaches 16.5%, both economically and statistically distinguishable from zero despite using a fraction of the full 47-signal set. A fourth specification, which selects the PLS component count to maximize test-period performance rather than fixing it *ex ante*, reaches 22.5%, close to the headline number but not treated as clean, since choosing components on the same window used to evaluate them contaminates the out-of-sample interpretation. A cleaner version of that exercise would choose components on a validation split carved out of the training period only, reserving the test window purely for final evaluation.

To be explicit about the procedure that rules out look-ahead bias, since this is the central credibility question for any 20,980-test screening exercise: for Specification C, both steps of the pipeline – (i) the univariate screening in Equation (1) that determines which signals enter the network, and (ii) the PLS estimation in Equation (2) that fits the propagation coefficients – are performed exactly once, using only the training window (May 1992–Feb 2015). The screening threshold ( $|\text{t-stat}(\hat{\gamma}_{j,s})| > 1.96$ , the standard two-sided 5% convention) is fixed *ex ante* and is not tuned to maximize either in-sample or out-of-sample fit; the same threshold is used throughout the paper, in sample and out of sample alike. Once the signal set and PLS loadings are fixed from the training window, the resulting model is applied mechanically to every month of the test window (Mar 2015–Dec 2024) to generate one-step-ahead forecasts: no re-estimation, no re-screening, and no test-period observation enters the selection of signals, the choice of threshold, or the estimation of coefficients at any point. Concretely, the sequence is: screen and estimate on May 1992–Feb 2015 → freeze the signal set and coefficients → forecast Mar 2015 using only information available through Feb 2015 → roll forward and repeat mechanically through Dec 2024 → compare the resulting forecasts to realized returns. This is also what separates Specification C from

Specification D in Table ??: in Specification D, only the PLS *component count* (not the signal set or coefficients) is chosen using the test window, which is precisely why Specification D is flagged as contaminated rather than reported as an equally valid robustness check.

One consequence of screening only on the training window is that the training-only screen finds 47 systematic signals for Specification C, fewer than the 65 systematic signals reported for the full 1992–2024 sample in Table 13. This is expected, not a discrepancy: with only 274 training months rather than 392, the screen has less power, so it is more conservative and retains fewer signals at the same 5% threshold. The 47-signal, training-only network is the one carried forward into Table ?? below for every sector’s out-of-sample evaluation; the 65-signal, full-sample network in Table 13 and Table 12 is used only for the in-sample analysis of Sections 3.1–3.2, and the two are never mixed within a single reported statistic.

This is also the specification carried forward into Table ?? below, which extends it to every sector.

Table ?? establishes the aggregate market result; Table ?? (Section 5) extends the same fixed-2-component PLS specification to every sector, decomposing each sector’s out-of-sample  $R^2$  into its systematic and idiosyncratic components. Manufacturing and Transportation show the strongest total out-of-sample fit (28.1% and 26.7%, respectively), consistent with their upstream position in the input-output network; Health and Education show negative total out-of-sample  $R^2$ , indicating that for these two sectors the network-based model does not outperform a constant-mean forecast out of sample. This heterogeneity, and the risk of overfitting in sectors with many idiosyncratic predictors relative to few observations, is revisited in the Conclusion.

This result rests on three pieces of evidence rather than any one alone, summarized in Table 1: the out-of-sample holdout itself, the permutation test in Section 6.2.1, and the cross-estimator consistency reported in Table 12.

### 3.4 Predictability from the Ten Strongest Signals per Sector

For each sector, we regress returns on the ten strongest LASSO-selected signals; full coefficient tables for the market and mining (the two sectors with the most contrasting signal composition) are in Table 15 and Table 16, and the same specification for all remaining sectors is summarized in this subsection rather than repeated sector-by-sector, since the economic pattern is what matters, not the individual predictor list.

**The aggregate market** is predicted at 15% by ten signals split five positive, five negative. The positive side is a demand-side story: employment in food manufacturing, hours in fruit and vegetable processing, industrial production of consumer goods, and aircraft parts

sales all track broad-based consumer and capital-investment demand. The negative side reflects cost pressure late in the business cycle. Hours in bakeries and waste collection, and employment in ambulance services, air transportation, and vocational rehabilitation, all proxy for rising labor costs in service-heavy industries. **Manufacturing** shares six of these same signals with the market, consistent with its high beta. **Mining** shares only two of ten – its predictors are almost entirely commodity- and construction-specific, reflecting its position as a raw-material supplier rather than a broad economic bellwether.

Across the remaining seventeen sectors, in-sample  $R^2$  from these ten-signal regressions ranges from 9% (Professional Services) to 18% (Information), and the same two patterns recur throughout. First, downstream, labor-intensive sectors (Retail, Construction, Accommodation, Arts) are predicted mostly by cost-side signals – rising hours or employment elsewhere in the economy forecasts *lower* returns for these sectors, consistent with the cost channel documented in Section 3.1. Second, sectors positioned as suppliers of physical inputs (Utilities, Real Estate, Wholesale) are predicted mostly by demand-side signals from the industries that consume their output – rising production or retail activity forecasts *higher* returns for these upstream suppliers, consistent with the demand channel. A small number of sectors (Healthcare, Finance, Administration) show a roughly even split between the two channels, which is itself informative. These are sectors whose customers include both households (cost-sensitive) and businesses (demand-sensitive), so no single channel dominates their return dynamics.

## 4 Conclusion

The results speak to how real economic information diffuses through the production network before it is fully reflected in prices, using the return predictability that diffusion leaves behind as the measurement device. Using a comprehensive dataset of 1,049 fundamental signals across NAICS industries, the results show that input-output fundamentals predict 40% of market return variation and, on average, 34% of variation across sectors, in sample. These findings are consistent with a meaningful share of monthly return variation reflecting real information still working its way through the network rather than already-priced news. The strongest evidence against a purely statistical, data-mined explanation is that employment and hours worked, two highly correlated measures of the same labor input, propagate with systematically opposite signs, exactly as predicted in advance by the demand-versus-cost distinction. A generic selection procedure has no reason to produce that pattern. Out of sample, evaluated on a 2015–2024 holdout never used in estimation, the preferred specification (47 systematic signals, PLS with 2 components fixed ex ante; Section 3.3) retains 23.8% predictive  $R^2$  for the aggregate market. Within that same 47-signal specification, in-sample fit is 39.3% (Table ??), so the relevant comparison is a decline from 39.3% to 23.8%, not from the larger 40% headline figure, which uses the full 65-signal network estimated on the complete sample rather than the smaller, training-only signal set used for the honest out-of-sample test (Section 3.3 explains why training-only screening finds fewer systematic signals). This decline is expected: the model is estimated on the training sample and evaluated on unseen data, so some attenuation from in-sample to out-of-sample fit is the normal signature of a genuine forecasting exercise rather than a sign of a flawed one; the fact that a substantial share of the in-sample fit survives suggests the diffusion structure generalizes rather than reflecting purely in-sample overfitting. The diffusion is far from uniform: upstream industries (manufacturing, mining) show the highest in-sample predictability (40–41%), consistent with information having more downstream links left to travel through, while downstream industries (arts, retail, construction) show lower in-sample predictability (28–32%), consistent with them sitting closer to the end of the chain where less remains to diffuse. Out of sample, the same upstream-downstream pattern holds but with a different sector ordering within the upstream group: Manufacturing and Transportation show the strongest total out-of-sample fit (28.1% and 26.7%, respectively; Table ??), with Mining close behind (21.6%).

The decomposition of propagation channels into systematic and sector-specific signals speaks to where in the network this diffusion is concentrated. Systematic channels align with CAPM betas: high-beta sectors have high systematic exposure, while low-beta sectors are driven by sector-specific fundamentals. This finding suggests that the diffusion process

operates through both market-wide channels and idiosyncratic, sector-specific ones simultaneously.

The empirical results are consistent with the theoretical distinction between demand and cost channels, though it is worth being explicit that these channels are inferred from the sign pattern of propagation coefficients rather than directly identified; the paper’s evidence establishes a systematic, theory-matching sign pattern, not a structural test that isolates demand from cost mechanisms. With that caveat, production and sales propagate positively across sectors, consistent with an aggregate demand channel. Inventories propagate negatively, consistent with a demand-deficiency signal. Human capital exhibits heterogeneous effects: employment and hours propagate positively in low labor-intensive industries (utilities, mining) but negatively in highly labor-intensive industries (retail), a pattern consistent with demand and cost channels operating differently depending on labor’s role in production. Extensive human capital (employment) also propagates more negatively than intensive human capital (hours), consistent with the higher fixed costs and longer-term commitments associated with hiring.

Two caveats qualify these results. First, sector-level out-of-sample performance is highly heterogeneous (Table ??): several sectors, particularly Health and Education, show negative out-of-sample  $R^2$ , and a deeper look at idiosyncratic signals alone (Appendix, Section 6.2.3) finds a similar pattern in Professional Services and Finance. The paper’s positive aggregate-market result should not be read as claiming uniform predictability across all 20 sectors. Second, the PLS component count in Specification D of Table ?? was chosen using the same test window used to evaluate performance, so that number is reported as an upper bound rather than a clean out-of-sample estimate; the headline specification (Specification C) fixes the component count ex ante precisely to avoid this issue. Third, the 20,980 univariate tests underlying the network raise a multiple-testing concern, which Section 6.2.1 addresses directly with a bootstrap null distribution constructed from the screening procedure itself.

Beyond the asset-pricing question this paper is built around, the propagation network suggests a possible practical use that this paper does not itself test. Employment, hours, production, sales, and inventories are released weeks before quarterly GDP and corporate earnings, so a network of the kind estimated here could, in principle, help a macro forecaster or sector analyst infer which parts of the economy a given month’s employment or production report is likely to reach next, ahead of those sectors’ own reporting. Whether the network actually improves on existing nowcasting approaches – which typically use factor models on a broader macro panel rather than an input-output structure – is a distinct empirical question, requiring a direct forecasting comparison against those benchmarks rather than an extension of the asset-pricing exercise here.

## 5 Tables and Figures

This section collects the tables and figures referenced in the main text, presented in order of first citation. Section 6.2 below is a separate appendix reporting additional robustness checks.

### 5.1 Tables

**Table 1:** Robustness at a Glance

| Check                                     |      | What it shows                                                                                                         | Status |
|-------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------|--------|
| Permutation test<br>(§6.2.1)              |      | Concentration of predictive power in broadly-propagating signals is essentially absent under a no-predictability null | Done   |
| Out-of-sample holdout                     |      | 23.8% $R^2$ for the market on 2015–2024 data never touched during estimation                                          | Done   |
| Cross-estimator consistency<br>(Table 12) | (Ta- | PLS, Ridge, LASSO, and Elastic Net give materially similar in-sample fit                                              | Done   |

OOS

**Table 2:** Out-of-Sample  $R^2$  Performance on Common Stable Signals (%)

| Sector        | Ridge       | Elastic Net | LASSO       | PLS 2       | PLS 3        | PLS 5        |
|---------------|-------------|-------------|-------------|-------------|--------------|--------------|
| Arts          | 13.90       | 11.39       | 9.33        | 11.81       | 5.14         | 2.04         |
| Finance       | 12.78       | 8.46        | 6.63        | 3.75        | 4.73         | 1.70         |
| Const         | 12.43       | 11.77       | 10.84       | 12.77       | 10.83        | 10.31        |
| Utils         | 11.87       | 11.53       | 11.07       | 1.04        | 7.52         | 6.39         |
| Educ          | 11.19       | 3.44        | -1.87       | -0.18       | -3.50        | -9.36        |
| Retail        | 10.76       | 8.85        | 6.71        | 7.29        | 7.86         | 3.57         |
| Info          | 10.01       | 5.10        | 2.14        | 1.81        | 0.33         | -1.47        |
| Market        | 8.81        | 6.84        | 5.78        | 4.38        | 5.05         | 3.07         |
| Others        | 7.87        | 4.85        | 2.90        | 5.44        | 3.00         | -1.99        |
| Manuf         | 7.71        | 6.01        | 5.04        | 4.09        | 3.65         | 0.59         |
| Mining        | 7.57        | 4.53        | 3.12        | 4.21        | 1.87         | 0.02         |
| Wholes        | 7.35        | 3.08        | 0.52        | 2.86        | -4.75        | -5.01        |
| Trans         | 7.34        | 5.53        | 4.91        | 5.69        | 3.76         | 2.98         |
| Admin         | 7.14        | 4.96        | 3.96        | 0.66        | 1.15         | 0.27         |
| Other Ser     | 6.51        | 1.52        | -0.72       | -2.22       | -3.55        | -6.87        |
| Accom         | 5.72        | 3.59        | 1.15        | -1.46       | -2.57        | -3.45        |
| Realest       | 3.14        | -0.40       | -2.49       | -1.20       | -3.84        | -4.53        |
| Agric         | 0.41        | -12.04      | -19.14      | -15.75      | -16.88       | -28.51       |
| Profser       | -2.86       | -10.14      | -12.82      | -18.38      | -20.99       | -22.96       |
| Health        | -3.43       | -7.53       | -11.08      | -18.92      | -19.78       | -22.46       |
| <b>Mean</b>   | <b>7.31</b> | <b>3.57</b> | <b>1.30</b> | <b>0.39</b> | <b>-1.05</b> | <b>-3.78</b> |
| <b>Median</b> | <b>7.64</b> | <b>4.91</b> | <b>3.01</b> | <b>2.33</b> | <b>1.51</b>  | <b>-0.73</b> |

Common Stable Signals = predictors significant in both training (1992-2014) and full sample (1992-2024). 70/30 train-test split. All methods use standardized predictors. Ridge uses all stable signals with L2 regularization.

LASSO and Elastic Net perform variable selection. PLS uses 2, 3, or 5 components for dimensionality reduction.

Negative values indicate worse performance than a naive historical mean forecast.

**Table 3: Industry Risk-Return Characteristics: Moments, CAPM, and Long-Run Performance**

| Industry  | N<br>Firms | Avg ME<br>(\$M) | M<br>Share<br>(%) | Mean<br>(%) | STD<br>(%) | Skew<br>(%) | Kurt | Beta | Alpha<br>(%) | $\alpha$ t-<br>stat | Sharpe | Buy &<br>Hold<br>(VW<br>Final<br>\$) |
|-----------|------------|-----------------|-------------------|-------------|------------|-------------|------|------|--------------|---------------------|--------|--------------------------------------|
| Retail    | 202        | 5.33E+11        | 4.69              | 1.10        | 5.61       | -0.12       | 1.78 | 1.01 | 0.16         | 1.28                | 0.131  | \$547.22                             |
| Accom     | 83         | 1.57E+11        | 1.38              | 1.08        | 6.38       | -0.49       | 2.31 | 1.11 | 0.08         | 0.53                | 0.111  | \$337.27                             |
| Agric     | 11         | 2.01E+10        | 0.18              | 1.02        | 6.45       | 0.12        | 1.39 | 0.82 | 0.19         | 0.91                | 0.101  | \$231.99                             |
| Others    | 318        | 2.11E+11        | 1.85              | 1.02        | 5.03       | -0.44       | 1.74 | 0.90 | 0.14         | 1.24                | 0.129  | \$1.04                               |
| Manuf     | 1719       | 2.54E+12        | 22.30             | 0.98        | 4.66       | -0.42       | 1.61 | 1.00 | 0.05         | 1.35                | 0.132  | \$345.48                             |
| Finance   | 677        | 9.31E+11        | 8.19              | 0.96        | 5.72       | -0.31       | 1.76 | 1.08 | -0.02        | -0.15               | 0.105  | \$211.70                             |
| Health    | 75         | 1.04E+12        | 9.11              | 0.96        | 6.77       | -0.18       | 2.77 | 1.04 | 0.00         | 0.03                | 0.088  | \$133.25                             |
| Admin     | 82         | 7.46E+10        | 0.66              | 0.97        | 5.91       | -0.10       | 1.38 | 1.12 | -0.03        | -0.29               | 0.102  | \$207.78                             |
| Info      | 360        | 3.35E+12        | 29.44             | 0.97        | 4.99       | -0.26       | 1.53 | 0.96 | 0.06         | 0.65                | 0.122  | \$293.84                             |
| Arts      | 26         | 3.56E+10        | 0.31              | 0.95        | 8.71       | 0.02        | 2.41 | 1.30 | -0.15        | -0.62               | 0.067  | \$46.15                              |
| Trans     | 95         | 3.18E+11        | 2.80              | 0.92        | 5.68       | -0.31       | 1.43 | 1.07 | -0.05        | -0.44               | 0.098  | \$165.28                             |
| Other_ser | 17         | 1.36E+11        | 1.20              | 0.90        | 6.68       | -0.33       | 2.36 | 1.07 | -0.08        | -0.43               | 0.079  | \$88.70                              |
| Educ      | 14         | 8.83E+11        | 7.76              | 0.90        | 8.30       | 0.33        | 5.36 | 1.05 | -0.06        | -0.25               | 0.064  | \$40.66                              |
| RealEst   | 81         | 4.36E+11        | 3.83              | 0.86        | 6.88       | -0.39       | 1.78 | 1.31 | -0.25        | -1.92               | 0.071  | \$62.21                              |
| Mining    | 182        | 1.76E+11        | 1.55              | 0.87        | 7.45       | -0.02       | 2.77 | 1.04 | -0.09        | -0.40               | 0.067  | \$51.58                              |
| Const     | 61         | 3.05E+10        | 0.27              | 0.83        | 7.91       | -0.11       | 1.37 | 1.39 | -0.33        | -1.78               | 0.058  | \$30.76                              |
| Wholes    | 145        | 1.40E+11        | 1.23              | 0.81        | 5.12       | -0.37       | 2.46 | 0.99 | -0.12        | -1.25               | 0.087  | \$93.56                              |
| Utils     | 137        | 2.60E+11        | 2.29              | 0.85        | 4.22       | -0.11       | 1.17 | 0.49 | 0.20         | 1.44                | 0.114  | \$159.76                             |
| ProfSer   | 156        | 1.12E+11        | 0.99              | 0.68        | 6.41       | -0.29       | 1.67 | 1.22 | -0.38        | -3.09               | 0.049  | \$23.92                              |
| Mean      | 244        | 6.02E+11        | 5.37              | 0.94        | 6.23       | -0.19       | 2.06 | 1.04 | -0.05        | -0.28               | 0.093  | \$181.99                             |
| Median    | 95         | 2.60E+11        | 2.29              | 0.94        | 6.38       | -0.26       | 1.76 | 1.05 | -0.05        | -0.29               | 0.094  | \$159.76                             |
| Max       | 1719       | 3.35E+12        | 29.44             | 1.10        | 8.71       | 0.33        | 5.36 | 1.39 | 0.20         | 1.44                | 0.132  | \$547.22                             |
| Min       | 11         | 2.01E+10        | 0.18              | 0.68        | 4.22       | -0.49       | 1.17 | 0.49 | -0.38        | -3.09               | 0.049  | \$1.04                               |

*Notes:* This table combines short-horizon risk-return characteristics with long-run performance. The “Buy & Hold (VW Final \$)” column reports terminal wealth (dollars) from investing \$1 in March 1992 and holding through October 2024 (32.7 years); VW = value-weighted final wealth.

**Table 4:** Correlation Matrix of Industry Returns (Value-Weighted, %)

| Industry  | Accom | Admin | Agric | Arts | Const | Educ | Finance | Health | Info | Manuf | Mining | Other | Others | ProfSer | RealEst | Retail | Trans | Utils | Wholes | Mean |
|-----------|-------|-------|-------|------|-------|------|---------|--------|------|-------|--------|-------|--------|---------|---------|--------|-------|-------|--------|------|
| Accom     | 75    |       |       |      |       |      |         |        |      |       |        |       |        |         |         |        |       |       |        | 69   |
| Admin     | 52    | 53    |       |      |       |      |         |        |      |       |        |       |        |         |         |        |       |       |        | 71   |
| Agric     | 72    | 65    | 47    |      |       |      |         |        |      |       |        |       |        |         |         |        |       |       |        | 50   |
| Arts      | 74    | 74    | 52    | 66   |       |      |         |        |      |       |        |       |        |         |         |        |       |       |        | 60   |
| Const     | 61    | 55    | 38    | 51   | 57    |      |         |        |      |       |        |       |        |         |         |        |       |       |        | 68   |
| Educ      | 75    | 74    | 55    | 66   | 75    | 55   |         |        |      |       |        |       |        |         |         |        |       |       |        | 52   |
| Finance   | 72    | 70    | 49    | 60   | 68    | 55   | 68      |        |      |       |        |       |        |         |         |        |       |       |        | 70   |
| Health    | 65    | 78    | 47    | 57   | 63    | 46   | 70      | 53     |      |       |        |       |        |         |         |        |       |       |        | 64   |
| Info      | 78    | 85    | 57    | 65   | 78    | 56   | 81      | 69     | 86   |       |        |       |        |         |         |        |       |       |        | 74   |
| Manuf     | 49    | 53    | 47    | 47   | 58    | 37   | 56      | 44     | 45   | 62    |        |       |        |         |         |        |       |       |        | 51   |
| Mining    | 71    | 74    | 50    | 62   | 70    | 53   | 70      | 68     | 59   | 70    | 46     |       |        |         |         |        |       |       |        | 64   |
| Other_ser | 71    | 76    | 52    | 60   | 69    | 53   | 75      | 65     | 65   | 80    | 58     | 69    |        |         |         |        |       |       |        | 67   |
| Others    | 72    | 80    | 51    | 62   | 73    | 58   | 71      | 62     | 82   | 87    | 53     | 64    | 70     |         |         |        |       |       |        | 68   |
| ProfSer   | 75    | 82    | 55    | 71   | 79    | 55   | 77      | 66     | 76   | 85    | 58     | 70    | 73     | 80      |         |        |       |       |        | 71   |
| RealEst   | 78    | 75    | 50    | 63   | 71    | 53   | 74      | 68     | 74   | 80    | 38     | 70    | 68     | 73      | 74      |        |       |       |        | 66   |
| Retail    | 76    | 77    | 55    | 65   | 78    | 55   | 80      | 66     | 68   | 84    | 65     | 70    | 79     | 74      | 80      | 71     |       |       |        | 71   |
| Trans     | 46    | 46    | 39    | 39   | 49    | 31   | 53      | 46     | 42   | 51    | 38     | 47    | 49     | 36      | 44      | 41     | 50    |       |        | 44   |
| Utils     | 77    | 81    | 58    | 69   | 80    | 60   | 79      | 73     | 71   | 87    | 64     | 74    | 78     | 78      | 83      | 74     | 84    | 52    |        | 73   |
| Wholes    |       |       |       |      |       |      |         |        |      |       |        |       |        |         |         |        |       |       |        |      |
| Mean      | 69    | 71    | 50    | 60   | 68    | 52   | 70      | 62     | 64   | 74    | 51     | 64    | 67     | 68      | 71      | 66     | 71    | 44    | 73     | 64   |

*Notes:* This table presents pairwise correlations of monthly value-weighted excess returns across 19 industries ( $\times 100$  for readability, so  $87 = 0.87$  correlation). The lower triangle shows correlations; the diagonal is omitted. The **Mean** row/column shows average correlation of each industry with all others. The overall mean correlation is 64%, indicating substantial comovement. Manufacturing-Wholesale (87%), Manufacturing-Professional Services (87%), and Information-Manufacturing (86%) show the strongest linkages. Utilities exhibits the weakest average correlation (44%), consistent with its regulated, defensive characteristics. Agriculture and Mining also show weak correlations (50–51%), reflecting commodity price drivers uncorrelated with broader equity markets.

**Table 5: Input-Output Structure: NAICS Industries**

| Seller<br>↓ to Buyer → | Agric | Mining | Utils | Const | Manuf | Wholes | Retail | Trans | Info | Finance | RealEst | ProfSer | Admin | Edu | Health | Arts | Accom | Others | HH | Govt | Exp |
|------------------------|-------|--------|-------|-------|-------|--------|--------|-------|------|---------|---------|---------|-------|-----|--------|------|-------|--------|----|------|-----|
| Agric                  | 2     | —      | —     | —     | 35    | 5      | 2      | —     | —    | —       | —       | —       | —     | —   | —      | —    | 1.5   | 1      | 25 | 10   | 5   |
| Mining                 | 1     | 5      | —     | 2     | 30    | 2      | 1      | 15    | 2    | 1       | —       | —       | —     | —   | —      | —    | —     | —      | 5  | 2    | 2   |
| Utils                  | 1     | 1      | 25    | 2     | 18    | 4      | 7      | 3     | 3    | 2       | 3       | 4       | 2     | 1   | 4      | 1    | 1     | 2      | 39 | 3    | —   |
| Const                  | 1     | 2      | 3     | 1     | 5     | 1      | 2      | 2     | 3    | 5       | 1.5     | 2       | 1     | 2   | 4      | 1    | 3     | 1      | 10 | 25   | 2   |
| Manuf                  | 8     | 5      | 2     | 15    | 25    | 20     | 15     | 10    | 5    | 3       | 2       | 3       | 5     | 1   | 8      | 2    | 10    | 1      | 35 | 15   | 5   |
| Wholes                 | 2     | 1      | 1     | 5     | 20    | 5      | 25     | 3     | 2    | 2       | 1       | 3       | 3     | 1   | 5      | 2    | 8     | 1      | 5  | 2    | 2   |
| Retail                 | —     | —      | —     | —     | 1     | —      | 1      | —     | —    | —       | —       | —       | —     | —   | —      | —    | —     | —      | 95 | 1    | —   |
| Trans                  | 3     | 5      | 1     | 2     | 20    | 15     | 10     | 5     | 3    | 3       | 2       | 5       | 5     | 1   | 3      | 1    | 1     | 1      | 8  | 2    | 2   |
| Info                   | 1     | 2      | 3     | 3     | 15    | 5      | 8      | 5     | 10   | 10      | 5       | 12      | 8     | 3   | 5      | 3    | 5     | 2      | 25 | 5    | 2   |
| Finance                | 2     | 3      | 5     | 10    | 5     | 8      | 5      | 8     | 8    | 8       | 1.5     | 8       | 8     | 2   | 8      | 2    | 5     | 1      | 30 | 10   | 3   |
| RealEst                | 1     | 2      | 3     | 8     | 3     | 10     | 3      | 3     | 5    | 8       | 5       | 10      | 5     | 3   | 5      | 2    | 8     | 2      | 40 | 8    | 2   |
| ProfSer                | 1     | 5      | 3     | 5     | 18    | 8      | 8      | 8     | 12   | 12      | 8       | 5       | 12    | 8   | 2      | 2    | 5     | 3      | 5  | 8    | 3   |
| Admin                  | 1     | 2      | 2     | 3     | 20    | 8      | 10     | 8     | 10   | 10      | 5       | 15      | 5     | 2   | 8      | 2    | 8     | 2      | 3  | 5    | 2   |
| Educ                   | —     | —      | —     | —     | 1     | —      | —      | —     | 1    | 1       | —       | 2       | 1     | 2   | 1      | 1    | —     | —      | 85 | 5    | —   |
| Health                 | —     | —      | —     | —     | 2     | —      | 1      | 1     | 1    | 2       | —       | 2       | 2     | 1   | 5      | 1    | 1     | 1      | 75 | 5    | —   |
| Arts                   | —     | —      | —     | —     | 1     | —      | 1      | —     | 2    | 1       | —       | 1       | 1     | 1   | 1      | 1    | 2     | —      | 85 | 3    | —   |
| Accom                  | —     | 1      | —     | 1     | 5     | 2      | 3      | 3     | 2    | 3       | 1       | 5       | 3     | 1   | 2      | 1    | 2     | 1      | 60 | 3    | —   |
| Other_ser              | 1     | 1      | 1     | 1     | 5     | 2      | 3      | 2     | 2    | 2       | 2       | 3       | 2     | 1   | 3      | 1    | 3     | 1      | 55 | 5    | 1   |
| Others                 | 2     | 3      | 2     | 3     | 15    | 8      | 10     | 8     | 8    | 10      | 5       | 10      | 8     | 2   | 8      | 3    | 8     | 2      | 20 | 10   | 5   |

*Notes:* This table presents a simplified Input-Output structure for 19 NAICS industries. Each row shows where that industry's output goes (i.e., who buys from them). Values represent percentages of the seller's total output. For example, Agriculture sells 35% of output to Manufacturing, 25% to Households, and 15% to Accommodation. Bold values ( $\geq 20\%$ ) indicate major flows. HH = Households (personal consumption), Govt = Government, Exp = Exports. The table reveals: (1) Consumer-facing industries (Retail, Education, Health, Arts) sell primarily to Households ( $> 70\%$ ); (2) B2B industries (Wholesale, Professional Services, Administrative) sell primarily to other businesses; (3) Manufacturing is the largest intermediate buyer, purchasing from most industries; (4) Utilities and Transportation serve both businesses and households. Data aggregated from BEA 2017 Benchmark Input-Output Use Tables to match my NAICS classification.

**Table 6:** Average Weekly Hours: Summary Statistics by Sector (1991–2025)

| <b>Sector</b>                   | <b>Vars</b> | <b>Mean</b> | <b>Std Dev</b> | <b>Range</b> |
|---------------------------------|-------------|-------------|----------------|--------------|
| Mining and Logging              | 9           | 44.7        | 3.5            | 27.8–53.8    |
| Manufacturing – Durable Goods   | 48          | 41.2        | 1.9            | 12.6–47.7    |
| Information                     | 5           | 42.2        | 1.0            | 38.9–46.1    |
| Manufacturing – Food/Beverage   | 15          | 40.7        | 2.0            | 32.6–49.0    |
| Manufacturing – Non-Durable     | 1           | 40.8        | 1.2            | 37.0–45.7    |
| Mining, Oil, and Gas Extraction | 1           | 40.2        | 0.6            | 37.2–41.7    |
| Construction                    | 33          | 38.9        | 3.0            | 29.0–52.2    |
| Retail – General                | 38          | 38.8        | 1.6            | 31.3–45.9    |
| Transportation & Warehousing    | 18          | 37.4        | 4.1            | 21.2–46.4    |
| Real Estate & Rental            | 34          | 36.3        | 2.3            | 26.9–46.4    |
| Finance & Insurance             | 15          | 35.7        | 3.2            | 22.2–47.5    |
| Professional Services           | 62          | 35.2        | 3.3            | 22.6–44.5    |
| Agriculture and Forestry        | 1           | 34.9        | 1.4            | 33.0–38.8    |
| Wholesale Trade                 | 1           | 34.8        | 1.7            | 32.7–39.8    |
| Construction (Aggregate)        | 1           | 33.3        | 1.4            | 31.9–37.6    |
| Other Services                  | 31          | 31.9        | 4.6            | 15.1–42.5    |
| Educational Services            | 43          | 31.8        | 2.8            | 17.3–40.7    |
| Retail – Specialized            | 43          | 31.4        | 4.2            | 19.1–41.8    |
| Health Care & Social Assistance | 27          | 26.5        | 3.6            | 15.3–38.5    |
| <b>Mean across all sectors</b>  | <b>22.4</b> | <b>36.7</b> | <b>2.5</b>     |              |

*Notes:* This table presents summary statistics for raw (untransformed) Average Weekly Hours data. Hours are measured per production/nonsupervisory employee. Manufacturing and mining sectors exhibit the highest average hours (40–45 hours/week), consistent with production-intensive operations. Service sectors show lower average hours (26–36 hours/week), reflecting part-time employment and institutional constraints. The sectoral differences in hours are structural and relatively stable over time.

**Table 7:** Overview of Datasets Used in Return Predictability Analysis

| Dataset                                 | Source          | Series       | Period             | Lag | Transformation    | Winsorization |
|-----------------------------------------|-----------------|--------------|--------------------|-----|-------------------|---------------|
| <b>Industry Returns</b>                 | CRSP            | 20           | 1992-03 to 2024-10 | 0   | Excess returns    | No            |
| <i><b>Labor Market Variables</b></i>    |                 |              |                    |     |                   |               |
| Employment (CES)                        | BLS             | 426          | 1992-03 to 2024-10 | 1   | Log differences   | 1%/99%        |
| Hours (CES)                             | BLS             | 426          | 1992-03 to 2024-10 | 1   | Log differences   | 1%/99%        |
| <i><b>Production Variables</b></i>      |                 |              |                    |     |                   |               |
| Industrial Production                   | Federal Reserve | 61           | 1972-01 to 2025-08 | 2   | Log differences   | 1%/99%        |
| <i><b>Manufacturing Variables</b></i>   |                 |              |                    |     |                   |               |
| Mfg Shipments                           | Census M3       | 20           | 1992-03 to 2025-07 | 2   | Already % change  | 1%/99%        |
| Mfg Inventories                         | Census M3       | 20           | 1992-03 to 2025-07 | 2   | Already % change  | 1%/99%        |
| <i><b>Wholesale Trade Variables</b></i> |                 |              |                    |     |                   |               |
| Wholesale Sales                         | Census WTSI     | 21           | 1992-01 to 2025-06 | 2   | Log differences   | 1%/99%        |
| Wholesale Inventories                   | Census WTSI     | 21           | 1992-01 to 2025-06 | 2   | Log differences   | 1%/99%        |
| <i><b>Retail Trade Variables</b></i>    |                 |              |                    |     |                   |               |
| Retail Sales                            | Census MRTS     | 36           | 1992-02 to 2025-06 | 2   | Already % change  | 1%/99%        |
| Retail Inventories                      | Census MRTS     | 9            | 1992-02 to 2025-06 | 2   | Already % change  | 1%/99%        |
| Retail I/S Ratio                        | Census MRTS     | 9            | 1992-02 to 2025-06 | 2   | First differences | 1%/99%        |
| <b>Total Predictors</b>                 |                 | <b>1,049</b> |                    |     |                   |               |

**Series:** Number of time series variables in each dataset.

**Period:** Sample period available for analysis (Year-Month format).

**Lag:** Number of months predictor is lagged relative to returns. Employment and Hours use 1-month lag due to timely release; all others use 2-month lag to account for data release schedules and market processing time.

**Transformation:** Method used to ensure stationarity. ‘Log differences’ =  $\Delta \log(x_t)$ ; ‘Already % change’ = data reported as month-over-month percentage changes; ‘First differences’ =  $\Delta x_t$ .

**Winsorization:** Outlier treatment by capping extreme values at 1st and 99th percentiles to reduce impact of data errors and extreme events.

**Sources:** BLS = Bureau of Labor Statistics; Census = U.S. Census Bureau; CRSP = Center for Research in Security Prices; Federal Reserve = Board of Governors of the Federal Reserve System.

**CES** = Current Employment Statistics; **M3** = Manufacturers’ Shipments, Inventories, and Orders; **WTSI** = Wholesale Trade Sales and Inventories; **MRTS** = Monthly Retail Trade Survey.

**Table 8:** Industry Factor Intensities

| <b>Industry</b>                                  | <b>Labor<br/>Intensity</b> | <b>Capital<br/>Intensity</b> | <b>Material<br/>Intensity</b> | <b>Energy<br/>Intensity</b> |
|--------------------------------------------------|----------------------------|------------------------------|-------------------------------|-----------------------------|
| Accommodation and Food Services                  | 0.91                       | 1.2                          | 0.65                          | 0.35                        |
| Administrative and Support Services              | 0.78                       | 0.4                          | 0.35                          | 0.15                        |
| Agriculture, Forestry, Fishing, and Hunting      | 0.66                       | 2.5                          | 0.55                          | 0.40                        |
| Arts, Entertainment, and Recreation              | 0.77                       | 3.5                          | 0.40                          | 0.20                        |
| Construction                                     | 0.75                       | 0.8                          | 0.68                          | 0.30                        |
| Educational Services                             | 0.81                       | 0.6                          | 0.25                          | 0.10                        |
| Finance and Insurance                            | 0.51                       | 0.7                          | 0.15                          | 0.05                        |
| Health Care and Social Assistance                | 0.85                       | 1.5                          | 0.45                          | 0.15                        |
| Information                                      | 0.48                       | 2.8                          | 0.40                          | 0.25                        |
| Manufacturing                                    | 0.44                       | 2.1                          | 0.72                          | 0.55                        |
| Mining, Quarrying, and Oil & Gas Extraction      | 0.37                       | 5.8                          | 0.45                          | 0.80                        |
| Other Services (Except Public Administration)    | 0.88                       | 0.3                          | 0.40                          | 0.20                        |
| Others (Conglomerates and Diversified)           | 0.50                       | 1.0                          | 0.60                          | 0.40                        |
| Professional, Scientific, and Technical Services | 0.57                       | 0.2                          | 0.30                          | 0.10                        |
| Real Estate and Rental and Leasing               | 0.54                       | 4.2                          | 0.20                          | 0.15                        |
| Retail Trade                                     | 0.83                       | 0.9                          | 0.70                          | 0.25                        |
| Transportation and Warehousing                   | 0.69                       | 4.5                          | 0.55                          | 0.70                        |
| Utilities                                        | 0.33                       | 6.5                          | 0.35                          | 0.95                        |
| Wholesale Trade                                  | 0.59                       | 0.5                          | 0.75                          | 0.30                        |
| <b>Min</b>                                       | 0.33                       | 0.2                          | 0.15                          | 0.05                        |
| <b>Max</b>                                       | 0.91                       | 6.5                          | 0.75                          | 0.95                        |
| <b>Mean</b>                                      | 0.66                       | 2.39                         | 0.47                          | 0.33                        |
| <b>Std</b>                                       | 0.15                       | 1.83                         | 0.18                          | 0.27                        |

*Notes:* This table presents factor intensities for 19 NAICS industries. **Labor Intensity** measures labor compensation as a share of total output (0 to 1 scale, from BEA Industry Accounts). **Capital Intensity** is the ratio of fixed capital stock to output (from BEA Fixed Assets). **Material Intensity** measures intermediate inputs as a share of gross output (from BEA Input-Output tables). **Energy Intensity** measures energy consumption per dollar of value added, normalized to 0–1 scale (from EIA energy surveys). Higher values indicate greater dependence on that factor. All measures represent 2000–2023 averages to capture structural industry characteristics.

**Table 9:** Univariate Regression Results Summary (Full Sample, 1992-2024)

|                                                         | <b>Total</b>  | <b>Significant</b> |            | <b>Positive</b> |             | <b>Negative</b>   |
|---------------------------------------------------------|---------------|--------------------|------------|-----------------|-------------|-------------------|
|                                                         | <b>Tests</b>  | <b>N</b>           | <b>%</b>   | <b>N</b>        | <b>%</b>    | <b>N (%)</b>      |
| <b>Panel A: Overall Statistics</b>                      |               |                    |            |                 |             |                   |
| All Predictors                                          | 20,980        | 1,360              | 6.5        | 711             | 52.3        | 649 (47.7)        |
| <b>Panel B: By Data Source</b>                          |               |                    |            |                 |             |                   |
| Employment (Extensive HC)                               | 8,520         | 556                | 6.5        | 242             | 43.5        | 314 (56.5)        |
| Hours (Intensive HC)                                    | 8,520         | 504                | 5.9        | 277             | 55.0        | 227 (45.0)        |
| Industrial Production                                   | 1,220         | 92                 | 7.5        | 69              | 75.0        | 23 (25.0)         |
| Sales (Mfg + Wholesale + Retail)                        | 1,540         | 110                | 7.1        | 103             | 93.6        | 7 (6.4)           |
| Inventories (Mfg + Wholesale + Retail, incl. I/S ratio) | 1,180         | 98                 | 8.3        | 20              | 20.4        | 78 (79.6)         |
| <b>Total (Panel B)</b>                                  | <b>20,980</b> | <b>1,360</b>       | <b>6.5</b> | <b>711</b>      | <b>52.3</b> | <b>649 (47.7)</b> |

**Total Tests:** Number of univariate regressions performed (20 sectors  $\times$  1,049 predictors = 20,980).

**Significant:** Predictors with p-value  $\leq 0.05$  using Newey-West HAC standard errors (6 lags).

**Positive/Negative:** Direction of significant coefficients. Percentages are of significant predictors.

**HC:** Human Capital. Employment measures extensive margin, Hours measure intensive margin.

**Categorization:** Sales aggregates manufacturing shipments (20 series), wholesale sales (21), and retail sales (36) = 77 signals  $\times$  20 sectors = 1,540 tests. Inventories aggregates manufacturing inventories (20), wholesale inventories (21), retail inventories (9), and the retail inventory-to-sales ratio (9) = 59 signals  $\times$  20 sectors = 1,180 tests. Together with Employment (426 signals), Hours (426), and Industrial Production (61), the five categories span the full 1,049-signal panel.

All regressions use appropriate lags: 1 month for employment/hours, 2 months for other predictors.

**Table 10:** Top 20 Signals by Signed Score

| Most Positive Signals                                                          |       | Most Negative Signals                                           |       |
|--------------------------------------------------------------------------------|-------|-----------------------------------------------------------------|-------|
| Signal                                                                         | Score | Signal                                                          | Score |
| HC_ex: Gift, novelty, and souvenir retailers                                   | +13   | HC_in: General warehousing and storage                          | -15   |
| HC_ex: Marketing research and public opinion polling                           | +13   | HC_ex: Air transportation                                       | -14   |
| HC_in: Framing contractors                                                     | +13   | HC_in: Professional employer organizations                      | -14   |
| HC_in: Other electrical equipment and component manufacturing                  | +11   | HC_ex: Structural steel and precast concrete contractors        | -13   |
| HC_in: Fruit and vegetable preserving and specialty food manufacturing         | +11   | HC_in: Warehousing and storage                                  | -13   |
| IP: Food/Beverage/Tobacco311312                                                | +11   | HC_in: Waste collection                                         | -13   |
| Wholesale Sales: hw                                                            | +11   | HC_in: Home health care services                                | -13   |
| HC_ex: Veterinary services                                                     | +10   | HC_in: Waste management and remediation services                | -12   |
| HC_ex: Corporate, subsidiary, and regional managing offices                    | +10   | HC_ex: Landscaping services                                     | -11   |
| HC_ex: Community food and housing, and emergency relief services               | +10   | HC_ex: Residential mental health and substance abuse facilities | -11   |
| HC_in: Construction and mining (except oil and gas) machinery manufacturing    | +10   | HC_in: Bakeries and tortilla manufacturing                      | -11   |
| HC_in: Transit and ground passenger transportation                             | +10   | Wholesale Inv: mng                                              | -11   |
| HC_ex: International, secondary market, and other investment activities        | +9    | HC_ex: Commercial banking                                       | -9    |
| HC_in: Engine, turbine, and power transmission equipment manufacturing         | +9    | HC_ex: Activities related to real estate                        | -9    |
| HC_in: Exterminating and pest control services                                 | +9    | HC_ex: Political and other similar organizations                | -9    |
| Mfg Sales: M3ADV                                                               | +9    | HC_in: Newspaper publishers                                     | -9    |
| IP: Consumer Value Added                                                       | +9    | HC_in: Offices of bank and other holding companies              | -9    |
| HC_in: Household and institutional furniture and kitchen cabinet manufacturing | +8    | HC_in: Golf courses and country clubs                           | -9    |
| Mfg Inv: M3ADV                                                                 | +8    | IP: Miscellaneous339                                            | -9    |
| HC_ex: Beer, wine, and liquor retailers                                        | +7    | HC_ex: Child and youth services                                 | -8    |

*Notes:* **Score:** Signed signal score = number of positive predictions minus number of negative predictions. **HC\_ex:** Human Capital Extensive margin (employment). **HC\_in:** Human Capital Intensive margin (hours worked). **IP:** Industrial Production. **Mfg:** Manufacturing Shipments or Inventories. **Wholesale/Retail:** Sales or Inventories in wholesale/retail trade. All signals shown have  $p$ -value  $\leq 0.05$  in univariate regressions with Newey–West HAC standard errors. The maximum absolute propagation score in the full panel is 15 (consistent with Figure 1 and Table 19).

**Table 11:** Univariate Predictability by Industry: Which Industries Are Most Predicted?

| Industry                      | Total Tests  | Significant (5%) | % Sig      | Mean R <sup>2</sup> (Sig) | Positive (%) | Negative (%) |
|-------------------------------|--------------|------------------|------------|---------------------------|--------------|--------------|
| Arts                          | 1,049        | 85               | 8.1        | 1.35                      | 15.3         | 84.7         |
| Others                        | 1,049        | 82               | 7.8        | 1.32                      | 78.0         | 22.0         |
| Utils                         | 1,049        | 81               | 7.7        | 1.46                      | 76.5         | 23.5         |
| Educ                          | 1,049        | 79               | 7.5        | 1.28                      | 30.4         | 69.6         |
| Retail                        | 1,049        | 72               | 6.9        | 1.30                      | 23.6         | 76.4         |
| Mining                        | 1,049        | 72               | 6.9        | 1.18                      | 63.9         | 36.1         |
| Trans                         | 1,049        | 70               | 6.7        | 1.23                      | 71.4         | 28.6         |
| Health                        | 1,049        | 68               | 6.5        | 1.29                      | 48.5         | 51.5         |
| Manuf                         | 1,049        | 67               | 6.4        | 1.26                      | 61.2         | 38.8         |
| Market                        | 1,049        | 65               | 6.2        | 1.23                      | 50.8         | 49.2         |
| Agric                         | 1,049        | 65               | 6.2        | 1.28                      | 72.3         | 27.7         |
| Const                         | 1,049        | 65               | 6.2        | 1.29                      | 27.7         | 72.3         |
| Profser                       | 1,049        | 64               | 6.1        | 1.12                      | 42.2         | 57.8         |
| Accom                         | 1,049        | 63               | 6.0        | 1.11                      | 41.3         | 58.7         |
| Admin                         | 1,049        | 62               | 5.9        | 1.21                      | 46.8         | 53.2         |
| Wholes                        | 1,049        | 62               | 5.9        | 1.30                      | 75.8         | 24.2         |
| Realest                       | 1,049        | 61               | 5.8        | 1.18                      | 67.2         | 32.8         |
| Finance                       | 1,049        | 60               | 5.7        | 1.32                      | 53.3         | 46.7         |
| Info                          | 1,049        | 60               | 5.7        | 1.36                      | 56.7         | 43.3         |
| Other Ser                     | 1,049        | 57               | 5.4        | 1.24                      | 47.4         | 52.6         |
| <b>Mean Across Industries</b> | <b>1,049</b> | <b>68.0</b>      | <b>6.5</b> | <b>1.27</b>               | <b>52.5</b>  | <b>47.5</b>  |

*Notes:* This table shows predictability for each industry, aggregating across all 1,049 predictor variables. % **Sig** = (Significant / 1,049)  $\times$  100; values exceeding 5% indicate above-random predictability. **Mean R<sup>2</sup> (Sig)** is the average R-squared for significant predictors (in percentage points), indicating economic magnitude. **Positive/Negative (%)** show coefficient sign distribution among significant predictors. Industries sorted by number of significant predictors.

**Table 12:** Sector Network Predictability: Full Sample In-Sample  $R^2$ 

| Sector      | N Pred.   | PLS         |          | Ridge       |          | LASSO       |           | Elastic Net |           |
|-------------|-----------|-------------|----------|-------------|----------|-------------|-----------|-------------|-----------|
|             |           | $R^2$       | Comp     | $R^2$       | Vars     | $R^2$       | Vars      | $R^2$       | Vars      |
| Manuf       | 67        | 40.5        | 2        | 40.8        | 67       | 43.0        | 57        | 40.5        | 63        |
| Market      | 65        | 39.7        | 2        | 37.9        | 65       | 41.5        | 56        | 38.9        | 63        |
| Mining      | 72        | 39.0        | 2        | 38.2        | 72       | 41.5        | 60        | 38.4        | 70        |
| Info        | 60        | 37.5        | 2        | 36.1        | 60       | 41.2        | 51        | 39.0        | 56        |
| Utils       | 81        | 36.1        | 2        | 39.3        | 81       | 39.7        | 53        | 38.8        | 73        |
| ProfSer     | 64        | 36.0        | 2        | 33.0        | 64       | 38.3        | 55        | 36.1        | 64        |
| Finance     | 60        | 35.0        | 2        | 33.9        | 60       | 36.4        | 44        | 33.8        | 56        |
| Health      | 68        | 34.9        | 2        | 34.4        | 68       | 38.1        | 48        | 36.0        | 61        |
| Admin       | 62        | 34.2        | 2        | 31.7        | 62       | 34.5        | 50        | 32.7        | 61        |
| Agric       | 65        | 33.4        | 2        | 31.8        | 65       | 34.6        | 50        | 33.1        | 58        |
| OtherSer    | 57        | 33.4        | 2        | 33.8        | 57       | 34.5        | 44        | 34.0        | 54        |
| Educ        | 79        | 32.8        | 2        | 33.2        | 79       | 36.9        | 46        | 32.9        | 67        |
| Trans       | 70        | 32.5        | 2        | 31.5        | 70       | 34.0        | 53        | 32.4        | 64        |
| Accom       | 63        | 32.4        | 2        | 32.4        | 63       | 34.9        | 50        | 33.6        | 59        |
| RealEst     | 61        | 32.0        | 2        | 30.4        | 61       | 34.2        | 53        | 31.6        | 60        |
| Wholes      | 62        | 31.9        | 2        | 29.1        | 62       | 33.9        | 50        | 31.6        | 56        |
| Const       | 65        | 31.0        | 2        | 28.4        | 65       | 31.4        | 51        | 29.6        | 62        |
| Retail      | 72        | 30.8        | 2        | 28.1        | 72       | 30.8        | 48        | 28.4        | 66        |
| Others      | 82        | 28.5        | 2        | 30.7        | 82       | 33.9        | 52        | 31.2        | 70        |
| Arts        | 85        | 27.9        | 2        | 28.6        | 85       | 32.0        | 51        | 29.8        | 74        |
| <b>Mean</b> | <b>68</b> | <b>34.0</b> | <b>2</b> | <b>33.2</b> | <b>–</b> | <b>36.3</b> | <b>51</b> | <b>34.1</b> | <b>63</b> |

**N Pred.:** Number of significant predictors in the sector network (univariate regressions,  $p \leq 0.05$ ).

**PLS:** Partial Least Squares with 2 components. **Comp** shows actual components used.

**Ridge:** Ridge regression with cross-validated  $\alpha$ . Uses all predictors.

**LASSO / Elastic Net:** Sparse methods with cross-validated tuning. **Vars** shows number of non-zero coefficients.

$R^2$ : In-sample fit (1992–2024) in percentage points. All predictors are standardized.

**Table 13:** Network Composition: Systematic vs. Sector-Specific Signals

| Sector        | Total<br>Signals | Systematic<br>(Market) | Sector-Specific Signals |           |          |          |          |           |
|---------------|------------------|------------------------|-------------------------|-----------|----------|----------|----------|-----------|
|               |                  |                        | Total                   | Unique    | +1 Sec   | +2 Sec   | +3 Sec   | +4 Sec    |
| Utils         | 81               | 9                      | 72                      | 39        | 10       | 8        | 5        | 10        |
| Educ          | 79               | 11                     | 68                      | 26        | 20       | 10       | 4        | 8         |
| Arts          | 85               | 16                     | 69                      | 18        | 20       | 15       | 4        | 12        |
| Agric         | 65               | 15                     | 50                      | 29        | 9        | 4        | 4        | 4         |
| Others        | 82               | 22                     | 60                      | 24        | 12       | 9        | 3        | 12        |
| Mining        | 72               | 21                     | 51                      | 17        | 13       | 8        | 3        | 10        |
| Const         | 65               | 20                     | 45                      | 10        | 15       | 5        | 4        | 11        |
| Health        | 68               | 21                     | 47                      | 16        | 11       | 7        | 5        | 8         |
| Retail        | 72               | 24                     | 48                      | 10        | 10       | 9        | 8        | 11        |
| Accom         | 63               | 23                     | 40                      | 10        | 6        | 10       | 3        | 11        |
| Wholes        | 62               | 27                     | 35                      | 6         | 6        | 7        | 2        | 14        |
| Other Ser     | 57               | 25                     | 32                      | 6         | 9        | 5        | 3        | 9         |
| Trans         | 70               | 32                     | 38                      | 6         | 10       | 7        | 3        | 12        |
| Admin         | 62               | 31                     | 31                      | 0         | 2        | 5        | 3        | 21        |
| Info          | 60               | 30                     | 30                      | 5         | 9        | 7        | 3        | 6         |
| Realest       | 61               | 31                     | 30                      | 5         | 6        | 4        | 3        | 12        |
| Profser       | 64               | 34                     | 30                      | 8         | 5        | 2        | 4        | 11        |
| Finance       | 60               | 32                     | 28                      | 7         | 5        | 2        | 6        | 8         |
| Manuf         | 67               | 49                     | 18                      | 0         | 2        | 2        | 6        | 8         |
| <i>Market</i> | 65               | 65                     | 0                       | 0         | 0        | 0        | 0        | 0         |
| <b>Mean</b>   | <b>68</b>        | <b>27</b>              | <b>41</b>               | <b>12</b> | <b>9</b> | <b>6</b> | <b>4</b> | <b>10</b> |

**Table 14:** Predictability of Systematic vs. Sector-Specific Signals

| Sector      | Signal Counts |           |           | PLS (2 Components) |             |             | LASSO           |                 |                 |
|-------------|---------------|-----------|-----------|--------------------|-------------|-------------|-----------------|-----------------|-----------------|
|             | All           | Sys       | SS        | All $R^2$          | Sys $R^2$   | SS $R^2$    | All $R^2(N)$    | Sys $R^2(N)$    | SS $R^2(N)$     |
| Manuf       | 67            | 49        | 18        | 40.5               | 36.9        | 11.8        | 43.0(57)        | 38.7(46)        | 12.5(18)        |
| Market      | 65            | 65        | 0         | 39.7               | 39.7        | 0.0         | 41.5(56)        | 41.5(56)        | 0.0(0)          |
| Mining      | 72            | 21        | 51        | 39.0               | 19.3        | 31.9        | 41.5(60)        | 19.2(20)        | 34.2(44)        |
| Info        | 60            | 30        | 30        | 37.5               | 29.3        | 21.1        | 41.2(51)        | 29.8(29)        | 22.8(27)        |
| Utils       | 81            | 9         | 72        | 36.1               | 11.0        | 35.6        | 39.7(53)        | 11.0(9)         | 39.6(49)        |
| Profser     | 64            | 34        | 30        | 36.0               | 25.3        | 23.1        | 38.3(55)        | 25.8(33)        | 23.8(26)        |
| Finance     | 60            | 32        | 28        | 35.0               | 28.4        | 20.2        | 36.4(44)        | 28.3(31)        | 19.2(27)        |
| Health      | 68            | 21        | 47        | 34.9               | 19.5        | 26.3        | 38.1(48)        | 19.8(20)        | 26.7(37)        |
| Admin       | 62            | 31        | 31        | 34.2               | 28.3        | 19.6        | 34.5(50)        | 28.3(30)        | 19.6(28)        |
| Agric       | 65            | 15        | 50        | 33.4               | 12.1        | 27.1        | 34.6(50)        | 12.1(14)        | 28.9(36)        |
| Other Ser   | 57            | 25        | 32        | 33.4               | 21.2        | 22.2        | 34.5(44)        | 21.1(22)        | 21.8(28)        |
| Educ        | 79            | 11        | 68        | 32.8               | 9.9         | 29.6        | 36.9(46)        | 10.5(11)        | 32.0(38)        |
| Trans       | 70            | 32        | 38        | 32.5               | 26.0        | 18.9        | 34.0(53)        | 26.3(31)        | 20.3(30)        |
| Accom       | 63            | 23        | 40        | 32.4               | 20.6        | 21.4        | 34.9(50)        | 20.9(23)        | 22.3(33)        |
| Realest     | 61            | 31        | 30        | 32.0               | 25.8        | 19.7        | 34.2(53)        | 26.0(30)        | 20.8(25)        |
| Wholes      | 62            | 27        | 35        | 31.9               | 22.1        | 20.2        | 33.9(50)        | 22.2(26)        | 20.7(30)        |
| Const       | 65            | 20        | 45        | 31.0               | 18.6        | 20.5        | 31.4(51)        | 18.8(20)        | 19.9(35)        |
| Retail      | 72            | 24        | 48        | 30.8               | 22.4        | 20.5        | 30.8(48)        | 22.6(24)        | 18.6(34)        |
| Others      | 82            | 22        | 60        | 28.5               | 22.1        | 19.6        | 33.9(52)        | 22.6(20)        | 23.5(35)        |
| Arts        | 85            | 16        | 69        | 27.9               | 17.1        | 21.1        | 32.0(51)        | 17.1(16)        | 23.2(37)        |
| <b>Mean</b> | <b>68</b>     | <b>27</b> | <b>41</b> | <b>34.0</b>        | <b>22.8</b> | <b>21.5</b> | <b>36.3(51)</b> | <b>23.1(26)</b> | <b>22.5(31)</b> |

**Table 15:** Market Return Predictability: Demand vs. Cost/Late-Cycle Signals

| Category                       | Variable                    | Coefficient | p-value |
|--------------------------------|-----------------------------|-------------|---------|
| <i>Demand Signals</i>          |                             |             |         |
|                                | Emp: Food Mfg               | +164.3***   | 0.002   |
|                                | Hours: Fruit/Veg Processing | +1.196***   | <0.001  |
|                                | IP: Consumer Value Added    | +17.8***    | 0.002   |
|                                | Mfg Sales: Aircraft/Parts   | +0.056***   | <0.001  |
|                                | Emp: Marketing Research     | +48.0***    | 0.003   |
| <i>Cost/Late-Cycle Signals</i> |                             |             |         |
|                                | Hours: Bakeries             | -1.506***   | 0.004   |
|                                | Hours: Waste Collection     | -2.676**    | 0.033   |
|                                | Emp: Ambulance Services     | -69.2***    | 0.009   |
|                                | Emp: Air Transportation     | -75.1**     | 0.015   |
|                                | Emp: Vocational Rehab       | -52.4***    | 0.008   |
| Observations                   |                             |             | 392     |
| Adjusted R <sup>2</sup>        |                             |             | 15%     |
| F-statistic                    |                             |             | 8.66*** |

**Variable Selection:** LASSO (CV) top 10, plus waste collection hours.

**Estimation:** Multivariate OLS, Newey-West HAC standard errors (6 lags), 1992-2024.

**Significance:** \*\*\* p<0.01, \*\* p<0.05, \* p<0.10.

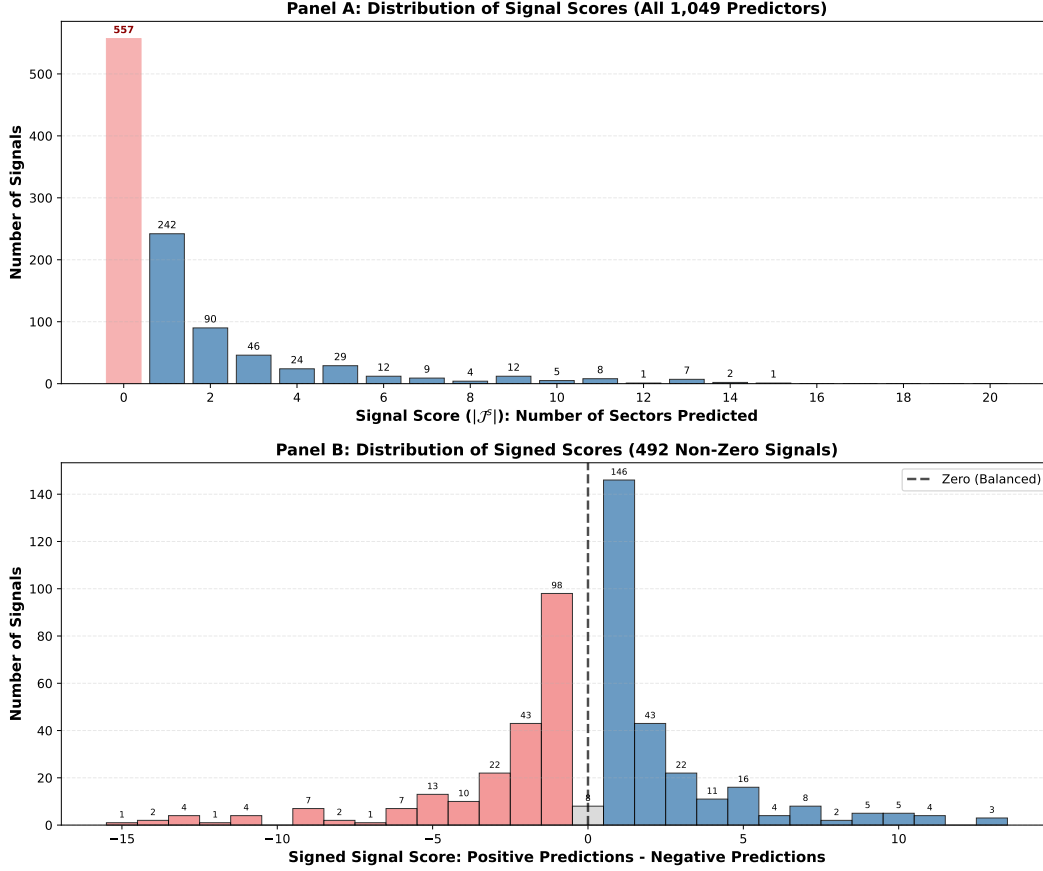
**Table 16:** Mining Return Predictability: LASSO-Selected Top 10 Predictors

| Category                   | Variable                           | Coefficient | p-value |
|----------------------------|------------------------------------|-------------|---------|
| <i>Negative Predictors</i> |                                    |             |         |
|                            | Emp: Social assistance             | -360.9**    | 0.022   |
|                            | Emp: Structural steel & precast... | -47.1***    | 0.004   |
|                            | Hrs: Golf courses & country clubs  | -3.567***   | 0.005   |
|                            | Hrs: Web search portals, libra...  | -2.247***   | 0.005   |
|                            | Hrs: General freight trucking,...  | -2.172**    | 0.014   |
| <i>Positive Predictors</i> |                                    |             |         |
|                            | Hrs: School & employee bus Tra...  | +1.153***   | 0.002   |
|                            | Hrs: Clothing, clothing access...  | +2.818**    | 0.040   |
|                            | Hrs: Household appliances, ele...  | +3.150***   | <0.001  |
|                            | IP: Computer and Electronics       | +85.7**     | 0.031   |
|                            | Emp: Beer, wine, & liquor reta...  | +187.8***   | <0.001  |
| Observations               |                                    |             | 392     |
| Adjusted R <sup>2</sup>    |                                    |             | 14%     |
| F-statistic                |                                    |             | 6.97*** |

**Variable Selection:** LASSO (CV) on 72 mining predictors, top 10 by coefficient magnitude.

**Signal Types:** 2 systematic (predict market), 8 [sector-specific](#) (idiosyncratic); sector-specific predictor names are shown in blue in the table above.

## 5.2 Figures



**Figure 1:** Distribution of Signal Scores in the Input–Output Propagation Network

*Notes:* **Panel A - Signal Score ( $|J^s|$ ):** Number of sectors that signal  $s$  significantly predicts ( $p$ -value  $\leq 0.05$ ). The red bar shows 557 signals (53.1%) with zero predictive power. **Panel B - Signed Score:** Net propagation direction for the 492 signals with non-zero scores. **Key Findings:** (1) 53.1% of signals have no predictive power; (2) Among predictive signals, 54.1% send only positive signals, 42.9% send only negative signals, and 3.0% send mixed signals; (3) The near 50–50 split between positive and negative propagation suggests a zero-sum mechanism in the short run. The full score-by-score counts (0 through 15) are reported in Table 19, together with the permutation-null benchmark used to test them.

## 5.3 Heatmaps

The previous sections described the distribution of signal scores and sector propagation scope. This section brings these together, first with a comprehensive heatmap showing the big picture of the propagation network across data sources and sectors, and then with each data source examined individually, starting with the extensive human capital to market value propagation network.

### 5.3.1 Comprehensive Heatmap

The comprehensive heatmap summarizes propagation channels for sectors across different data classes. Rows represent sectors including the market portfolio, while columns represent different datasets. Cell values show positive (numerator) and negative (denominator) propagation coefficients. Blue represents positive coefficients and red represents negative coefficients.

**Comprehensive Predictability Map: All Data Sources**  
(Cell format: Positive/Negative significant predictors)

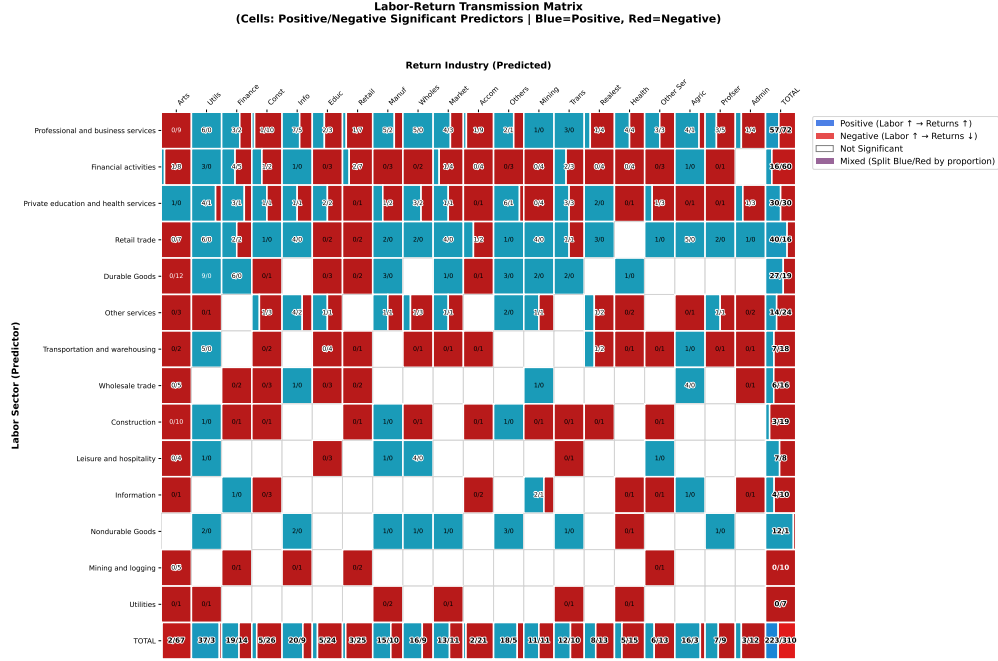
|           | Emp.<br>(Ext.) | Emp.<br>(Int.) | Prod.<br>(IP) | Inv.<br>Mfg | Inv.<br>Whsl | Inv.<br>Retail | Sales<br>Mfg | Sales<br>Whsl | Sales<br>Retail | TOTAL   |
|-----------|----------------|----------------|---------------|-------------|--------------|----------------|--------------|---------------|-----------------|---------|
| Others    | 14/6           | 19/5           | 13            |             | 3            |                | 8            | 1             | 1               | 56/14   |
| Utils     | 34/7           | 11/5           | 5             |             | 1            | 1              |              | 1             | 7/1             | 59/15   |
| Agric     | 19/4           | 11/8           | 3             |             |              | 4              | 2            | 1             | 2               | 42/12   |
| Wholes    | 15/10          | 21/7           | 1             | 1           | 1            | 1              |              | 2/1           | 2               | 43/19   |
| Trans     | 7/9            | 16/6           | 3             | 1           | 2            | 1              | 1            | 1             | 1               | 30/18   |
| Mining    | 8/14           | 21/12          | 4             |             |              |                | 1            |               | 2               | 36/26   |
| Realest   | 8/13           | 15/7           | 4             |             | 1            |                | 1            | 1             |                 | 29/21   |
| Manuf     | 11/13          | 13/10          | 4/1           | 1           | 1            |                | 3            | 1             | 1               | 34/25   |
| Info      | 12/11          | 17/14          | 3/1           | 1           | 2            |                | 1            |               | 3               | 37/28   |
| Market    | 9/10           | 10/9           | 3/1           | 1           | 2            |                | 1            | 1             | 1               | 26/22   |
| Finance   | 13/10          | 10/12          |               | 1           | 2            |                |              | 2             | 2/1             | 27/26   |
| Other Ser | 10/11          | 8/10           | 2/1           | 1           | 4            |                | 1            | 2             | 1/1             | 24/28   |
| Accom     | 5/14           | 11/8           | 2             | 2           | 1/1          |                |              | 1             | 1/1             | 21/26   |
| Admin     | 4/17           | 12/9           | 4/2           |             | 1            | 1              |              | 1             | 3               | 24/30   |
| Health    | 5/18           | 7/16           | 7             |             | 1            | 3              |              | 7             | 4/1             | 30/39   |
| Profser   | 7/12           | 3/20           | 5/1           |             | 1            |                | 3            | 1             | 3               | 26/35   |
| Const     | 4/19           | 8/8            | 1/2           | 1           | 1            | 2              |              | 1             | 1/3             | 16/35   |
| Educ      | 3/22           | 12/13          | 6             | 9           |              | 2              | 1            | 1             | 1               | 16/54   |
| Retail    | 3/25           | 8/11           | 4             | 1/3         | 7            | 6              |              | 1             |                 | 12/57   |
| Arts      | 1/34           | 5/16           | 3             | 6           | 4            | 1              |              |               | 2/1             | 11/62   |
| TOTAL     | 191/279        | 243/207        | 67/19         | 7/22        | 2/34         | 5/17           | 23           | 23/4          | 37/10           | 599/592 |

|                                                                                                                                |                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <span style="background-color: #007bff; color: white; padding: 2px 5px;"> </span> Positive (predictor growth → higher returns) | <span style="background-color: #f8d7da; color: #85c1e9; padding: 2px 5px;"> </span> No significant predictors               |
| <span style="background-color: #dc3545; color: white; padding: 2px 5px;"> </span> Negative (predictor growth → lower returns)  | <span style="background-color: #d6d8db; color: #6c757d; padding: 2px 5px;"> </span> Mixed (split cell: blue left/red right) |

**Figure 2: Comprehensive Industry-Level Information Transmission Matrix**

The TOTAL row reveals systematic patterns across information categories. The extensive margin shows 192 positive versus 279 negative relationships (41% positive). The intensive margin shows 243 positive versus 207 negative relationships (54% positive). Combined, labor overall remains moderately negative-dominant at 47% positive.

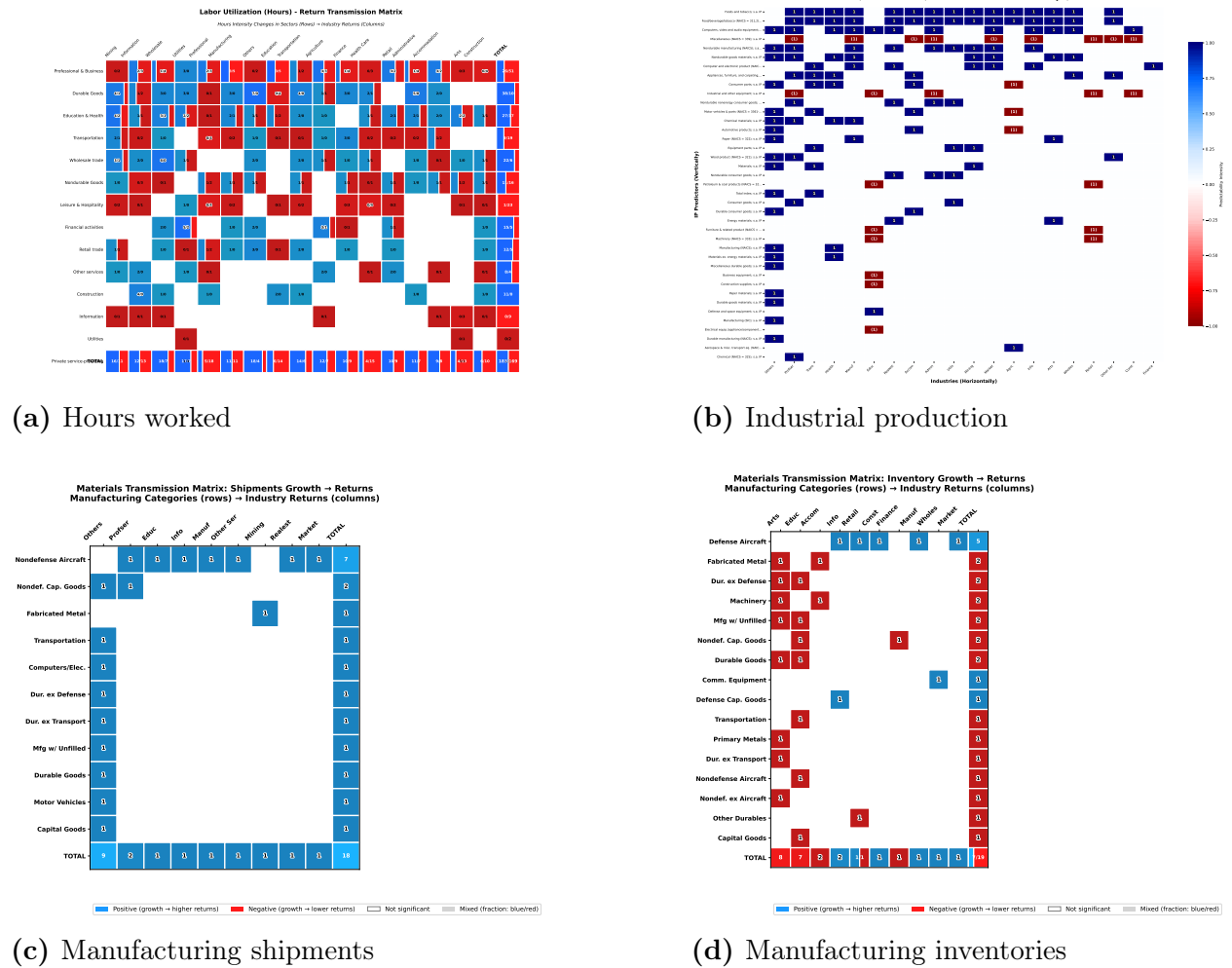
### 5.3.2 Employment-Return Propagation Network



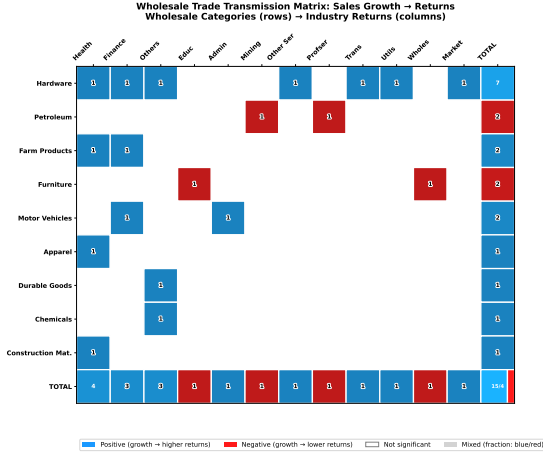
**Figure 3:** Employment-Return Propagation Network. This heatmap visualizes how employment changes in 19 CES supersectors (rows/left) predict returns across 20 industries (columns/top). Each cell displays “Positive/Negative” counts of significant predictors (5% level, Newey-West HAC SE 6 lags).

### 5.3.3 Remaining Data Sources: Hours, Production, Manufacturing, Wholesale, and Retail

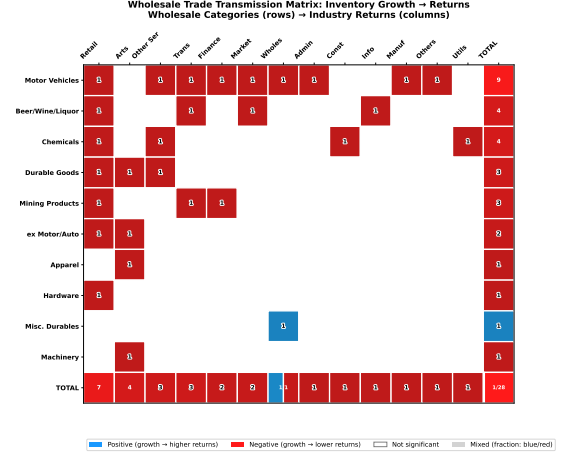
The remaining eight propagation heatmaps – hours worked, industrial production, manufacturing shipments and inventories, wholesale sales and inventories, and retail sales and inventories – follow the same construction as Figure 3 (rows are origin industries/data sources, columns are destination sectors, cells show positive/negative counts of significant predictors at the 5% level). They are collected together in Figures 4 and 5 rather than as eight separate full pages, since the reading convention is identical across all eight; the hours heatmap is directly comparable to Figure 3 and is discussed in that comparison in Section 3.1.



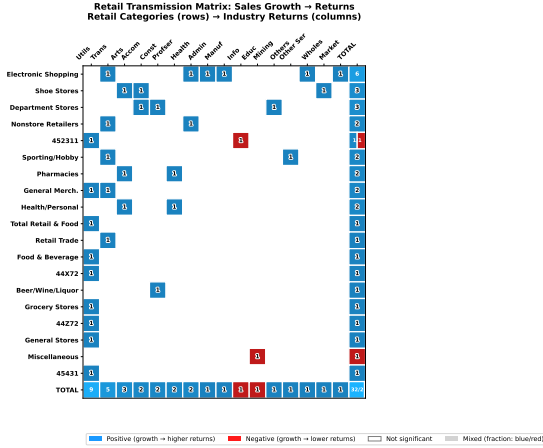
**Figure 4:** Propagation Network Heatmaps: Hours, Production, and Manufacturing. Same reading convention as Figure 3: rows are origin data sources, columns are destination sectors, cells show positive/negative counts of significant predictors (5% level).



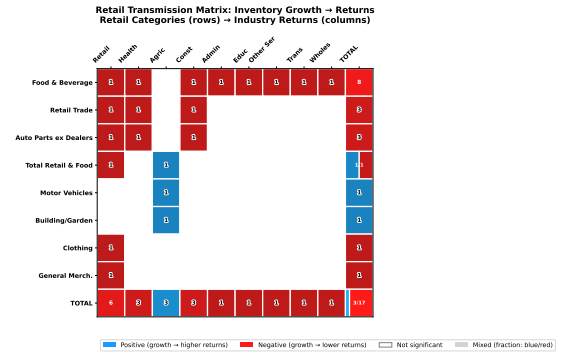
(a) Wholesale sales



(b) Wholesale inventories



(c) Retail sales



(d) Retail inventories

**Figure 5:** Propagation Network Heatmaps: Wholesale and Retail. Same reading convention as Figure 3: rows are origin data sources, columns are destination sectors, cells show positive/negative counts of significant predictors (5% level).

## 6 Appendix

This appendix has three parts. Section 6.1 provides the full data construction detail (classification choices, transformations, lag structures, and source-by-source summary statistics) that is condensed to brief summaries in Section 1 of the main text. Sections 6.2.1 and 6.2.2 report the identification checks that support the main results in Section 3: Section 6.2.1 implements a permutation test of the entire network-construction procedure against a no-predictability null, and Section 6.2.2 discusses further robustness dimensions – recursive and expanding-window re-estimation, a random-network placebo, stability across NBER recession and expansion subsamples, crisis-exclusion tests, pre/post-2000 subsamples, equal-

versus value-weighted returns, Fama–French industry portfolios, and alternative lag structures – that would further isolate the reported predictability from a single sample period, weighting scheme, classification system, or lag choice. Section 6.2.3 summarizes how well sector-specific (idiosyncratic) signals hold up out of sample once systematic signals are accounted for separately.

## 6.1 Detailed Data Construction

This subsection provides the full construction detail, classification discussion, and source-by-source summary statistics that Section 1 of the main text summarizes briefly.

### 6.1.1 Industry Returns: Detailed Construction

**Industry Classification** The first step in understanding industries requires a specific definition of what constitutes an industry. In defining industries, there is substantial ambiguity among different agents. For example, government agencies, whose main goal is collecting information on units of the economy and reporting them, use two main classifications: SIC and NAICS, with the latter created in 1997 as a response to inefficiencies in the former. NAICS classification is hierarchical, with 20 main sectors, where each sector is classified into subsectors and then industries; of these 20 official NAICS sectors, this paper constructs return series for 19 (Public Administration is excluded, since it has no publicly traded equity and therefore no observable market return), plus the aggregate value-weighted market portfolio, yielding the 20 return series (19 sectors + market) used consistently throughout the empirical analysis. Therefore, in this classification, sectors and industries represent two different units. On the other hand, practitioners such as mutual funds, whose main goal is facilitating sector investment, developed the Global Industry Classification Standard (GICS). Developed in 1999 by MSCI and Standard & Poor’s, GICS classifies publicly traded firms into 11 sectors, 25 industry groups, 74 industries, and 163 sub-industries. The sector ETFs are available for these 11 sectors only.<sup>2</sup>

Finally, academics whose main concern is understanding movements in industries, such as Fama and French (1997), developed several industry groupings (FF12 to FF49) by mapping firms’ SIC codes into economically interpretable industries. These classifications are designed for asset pricing research, analyzing return comovement, and studying factor structures in financial markets. The grouping prioritizes statistical similarity in returns over production characteristics (Fama & French, 1997). There are also Text-Based Classification systems

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<sup>2</sup>A companion chapter examines institutional holdings, their industry composition, and the market values of sector ETFs in greater detail.

such as the Hoberg–Phillips framework. This modern approach constructs industry groups using textual similarity of firms’ product descriptions in SEC 10-K filings (Hoberg & Phillips, 2016).

This research uses the NAICS classification system; therefore, sectors refer to the 20 main sectors in this classification, and industry refers to the NAICS 3-digit and 4-digit industries. NAICS is used in preference to other classifications for two reasons. First, NAICS classifies industries based on similarity in their production processes among firms, which is necessary to build a propagation network grounded in input-output relationships. Second, almost all industry-level information collected by government agencies is based on the NAICS classification system, as detailed further below.

**Return Construction** Using CRSP monthly data, firm-level information on prices is collected and aggregated based on the NAICS classification to derive industry-level return characteristics<sup>3</sup>. Monthly returns on individual firms in the U.S. are from CRSP, then aggregated according to NAICS classification codes to create returns on monthly sectors. NAICS classifies industries based on a hierarchical strategy which starts from 19 main sectors and then subsectors and industries. This study focuses on predictability at the sector level. There are 19 sectors and one aggregate market portfolio, value-weighted following standard Fama-French construction. The sample covers the period from March 1992 to October 2024, providing 392 monthly observations for each sector.

Table 3 provides an overview of industries’ market value, risk–return moments, and systematic risk. Industries are sorted by their mean returns.

The first observation concerns the substantial heterogeneity in both the number of firms and market shares across sectors. Some sectors are considerably larger than others. The *Information* sector is the largest, accounting for roughly 30% of total market capitalization, followed by *Manufacturing*, which represents about 22%. Together, these two sectors comprise roughly half of the entire market. In contrast, sectors such as *Agriculture*, *Arts*, *Construction*, *Administration*, and *Professional Services* each account for less than 1% of total market value.<sup>4</sup>

The second observation relates to industry performance, as captured by the moments of their return distributions. Sectors are ordered by their mean returns. The *Retail* industry, followed by *Accommodation* and *Agriculture*, exhibits the highest mean returns. In contrast, *Professional Services*, *Utilities*, *Wholesale*, and *Construction* have experienced the lowest average returns over the past three decades. Return volatility also varies substantially: sec-

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<sup>3</sup>A comprehensive data appendix elaborates further on data sources and construction.

<sup>4</sup>Future work could further disaggregate these smaller sectors into subsectors to achieve a more balanced comparison across industries.

tors such as *Utilities*, *Manufacturing*, and *Information* display relatively stable performance, whereas *Mining*, *Construction*, *Education*, and *Arts* exhibit considerably higher volatility.

The heterogeneity extends to higher moments. Skewness varies from 0.33 for *Education* to -0.49 for *Accommodation*. Only three industries—*Education*, *Arts*, and *Agriculture*—exhibit positive skewness, suggesting infrequent but large positive return realizations. The first two are relatively small sectors, so their skewness may be driven by idiosyncratic firm-level events. However, *Education*, representing nearly 8% of market value, is a large sector; its positive skewness likely reflects structural developments, such as the early-2000s rise of for-profit education providers like Grand Canyon Education. In contrast, industries such as *Accommodation* and *Manufacturing* exhibit pronounced negative skewness, indicating occasional sharp declines amid generally stable performance.

Kurtosis also differs notably across sectors. Except for *Education*—which displays strong fat tails (kurtosis = 5.36)—most sectors show relatively thin tails, implying fewer extreme return events. The industries with the thinnest tails include *Utilities*, *Construction*, and *Transportation*.

Systematic risk, measured by CAPM betas, shows substantial variation as well. Betas range from 0.49 for the *Utilities* sector—indicating limited sensitivity to market-wide fluctuations—to 1.39 for *Construction*, which is highly exposed to aggregate economic shocks. Other low-beta sectors include *Agriculture* (0.82), while other high-beta sectors include *Real Estate* (1.31) and *Arts* (1.30).

After controlling for systematic risk, abnormal returns (alphas) also vary significantly, ranging from -0.38% to 0.20%. The *Professional Services* sector earns the most negative alpha, followed by *Construction* and *Real Estate*, whereas *Utilities*, *Agriculture*, and *Retail* deliver the highest positive alphas—though only three are statistically significant.

Finally, the Sharpe ratio, a measure of risk-adjusted performance, ranges from 0.05 to about 0.13 across sectors. The last column of Table 3 reports the terminal value of a value-weighted buy-and-hold strategy initiated at the start of the sample. Excluding the residual “Others” category, *Professional Services* has the lowest terminal value, while the *Retail* sector achieves the highest, growing an initial \$1 investment to approximately \$547 by 2024.

**Correlation Structure** Different industries are interconnected through various economic relationships. Some industries serve as customers for others, some supply key production inputs, and others operate at different stages of the same supply chain. For example, *Mining*, *Construction*, and *Utilities* are closely linked through input-output dependencies. Such interconnections naturally generate correlations in industry performance. To provide an initial view of these relationships, Table 4 presents the correlation matrix of monthly industry

returns. The average pairwise correlation across all sectors is about 64%. The last row of the table reports each sector’s mean correlation with all others. As expected, *Utilities* exhibits the lowest average correlation, while *Manufacturing*, *Wholesale*, *Transportation*, *Real Estate*, and *Administrative Services* display the highest.

Within-industry correlations reveal interesting patterns. *Manufacturing* shows the strongest average correlation (74%), with particularly high comovement with *Wholesale* (87%), *Professional Services* (87%), *Information* (86%), *Real Estate* (85%), and *Administrative Services* (85%). This reflects *Manufacturing*’s central role in the U.S. economy—as both a demander of business services (*Professional* and *Administrative*) and a supplier of goods to distributors (*Wholesale*) and technology-related sectors (*Information*). The next section explores these inter-industry linkages in more detail.

In contrast, *Utilities* exhibits the weakest average correlation (44%), making it an effective diversifier in a portfolio context. Its highest correlation is only 53% (with *Finance*), while correlations with *Education* (31%), *Professional Services* (36%), and *Agriculture* (39%) remain below 40%. *Wholesale* and *Transportation* follow *Manufacturing* as the next two sectors with the strongest overall comovement.

Finally, *Agriculture* (50%) and *Mining* (51%) exhibit below-average correlations with other sectors, consistent with their positions early in the production cycle. As primary producers of raw materials, their performance is less synchronized with downstream industries.

**Input-Output Table** To understand industries and their network, this section briefly describes where each industry sells its outputs and who supplies its inputs. This Input-Output table was constructed by aggregating data from the Bureau of Economic Analysis (BEA) 2017 Benchmark Input-Output Use Tables, which report the dollar value of commodities purchased by each industry from every other industry. The BEA compiles these tables every five years using data from the Economic Census, Annual Survey of Manufactures, Service Annual Survey, and IRS tax records, surveying businesses about their expenditures on inputs from different sectors. The detailed Use table (available at <https://www.bea.gov/data/industries/input-output-accounts-data>) contains 400+ industries at the 6-digit NAICS level; these were aggregated up to the 19 broader 2-digit NAICS categories used here by summing all purchases within each group. For each seller industry (row), the percentage of total output going to each buyer industry (column) was calculated by dividing the dollar flows by the seller’s gross output, resulting in percentage shares that show the customer composition.

Based on input-output tables, Agriculture purchases from manufacturing, transportation, and other industries and sells predominantly to manufacturing, households, government, and

accommodation. Mining sells predominantly to utilities and manufacturing. Utilities sells to manufacturing, households, and all other industries. The construction industry sells to government, households, and real estate. Manufacturing sells to all other industries, with other manufacturing firms, wholesalers, households, and government being the dominant customers. Wholesalers purchase predominantly from manufacturing, use transportation as an input, and sell predominantly to manufacturing and retailers. Retailers purchase from manufacturing, wholesalers, and some other industries but sell to households solely. Transportation sells to manufacturing, wholesale, and retail for transportation of goods and services, and to households for passenger transportation. Information sells to a variety of industries, households, manufacturing, and services. Finance sells to households, government, real estate, and manufacturing. Real estate sells to households, retail, and services. Professional services sells to manufacturing, retail, and some other industries. Education, health, arts, accommodation, and other services sell predominantly to households. Finally, other industry sells to a variety of industries and households.

### 6.1.2 Human Capital: Detailed Construction

The second data type is industry-level human capital from Current Employment Statistics (CES). The Current Employment Statistics program is a monthly survey of approximately 121,000 businesses and government agencies, covering roughly one-third of all nonfarm payroll jobs in the United States. The sample is drawn from unemployment insurance (UI) tax records and represents a comprehensive snapshot of employment conditions across nonagricultural establishments. CES provides detailed, industry-level estimates of number of employees and average weekly hours worked by employees in 426 industries. The data are released monthly, on the first Friday of the following month, making CES one of the earliest official indicators of current economic activity.

CES data capture two dimensions of human capital: *extensive margin* (employment levels) and *intensive margin* (hours worked). The extensive margin measures the number of production and nonsupervisory employees, reflecting hiring and firing decisions that signal long-term labor demand and cost commitments. The intensive margin measures average weekly hours worked per employee, reflecting near-term production adjustments and labor utilization intensity. This distinction is theoretically important: extensive margin adjustments (hiring) entail higher fixed costs and signal longer-term commitments, while intensive margin adjustments (hours) are more flexible and reflect near-term production needs (Hamer-mesh, 1993). As shown below, this distinction manifests in differential propagation patterns: extensive human capital propagates more negatively than intensive human capital, consistent with cost channel predictions.

**Extensive Margin: Employment** The employment data are available from 1960 for aggregate sectors; however, for detailed industries, the variables are mostly available from January 1991 to July 2025, which includes 415 months. The following table shows the number of sectors for which the data are available, which are 19.

The following table gives a big picture of number of production and supervisory employees per sectors. All in all, on average in each month there are 92 million employees in production and nonsupervisory positions. The employees in total private economy is 92 million monthly. About 77 million of them are employed in service industry and 16 million are employed in goods producing industries. Transportation, education and health services, retail and manufacturing are the main labor intensive industries.

The next columns show the same for stationary transformed data. Total private has an average monthly growth of 0.13 percent per month, with a std of 0.2. On average, services have had higher growth but goods producing have not. Before transformation, the raw employment data reveal substantial sectoral heterogeneity. Service sectors demonstrate robust employment growth—Educational Services (+143%), Transportation & Warehousing (+108%), and Professional Services (+104%)—while manufacturing sectors experienced significant declines: Non-Durable Manufacturing (-28%), Durable Goods Manufacturing (-23%), and Information (-22%). This pattern reflects the U.S. economy’s structural transformation from manufacturing to services over the past three decades.

Employment levels also vary dramatically in scale. Large aggregate sectors like Total Private (92,366K average), Construction (76,717K), and Wholesale Trade (21,701K) contrast with granular subsectors like Mining & Logging (499K average) and Information (2,244K). This heterogeneity motivates the use of standardized transformations (log differences) that make growth rates comparable across sectors regardless of absolute employment levels.

To ensure stationarity for time series analysis, the raw employment levels are transformed into log differences:  $\Delta \log(x_t) = \log(x_t) - \log(x_{t-1})$ . This transformation converts the data into month-over-month growth rates, which can be interpreted as percentage changes in employment. The data are winsorized at the 1st and 99th percentiles to mitigate the influence of extreme outliers. Following the literature on macroeconomic predictors and accounting for the timely release of CES data, a 1-month lag structure is used when employment variables predict industry returns.

**Intensive Margin: Average Weekly Hours** The CES Average Weekly Hours data reveal substantial variation in labor utilization across sectors. Table 6 presents summary statistics. Manufacturing and mining sectors exhibit the highest average hours (40–45 hours per week), consistent with production-intensive operations requiring extended work sched-

ules. Service sectors show significantly lower average hours (26–36 hours per week), reflecting part-time employment patterns, flexible scheduling, and institutional constraints.

The stability of weekly hours across time (modest standard deviations) contrasts with the trending behavior of employment levels, confirming the appropriateness of different transformation strategies. Hours fluctuate around structural means, making first differences sufficient for stationarity, while employment requires more aggressive detrending via log differences.

To ensure stationarity, the raw hours data are transformed into first differences:  $\Delta x_t = x_t - x_{t-1}$ . This transformation captures changes in hours worked, which fluctuate around structural means. The data are winsorized at the 1st and 99th percentiles to mitigate extreme outliers. As with employment, a 1-month lag structure is used for hours variables, reflecting the timely release of CES data.

### 6.1.3 Industrial Production: Detailed Construction

The Industrial Production (IP) dataset comprises 61 seasonally adjusted series measuring physical output across three main sectors: manufacturing (durable and non-durable goods), mining, and utilities. Sourced from the Federal Reserve Economic Data (FRED), the data spans January 1972 to August 2025, providing 644 monthly observations. Each series is reported as an index with base year 2017=100, where values represent the level of physical production relative to the 2017 benchmark. The index is constructed using the Fisher ideal formula, which aggregates production quantities weighted by value-added shares, capturing both changes in output volumes and shifts in the industrial composition over time.

To ensure stationarity for time series analysis, the raw index levels are transformed into log differences:  $\Delta \log(x_t) = \log(x_t) - \log(x_{t-1})$ . This transformation converts the data into month-over-month growth rates, which can be interpreted as percentage changes in production. The data are winsorized at the 1st and 99th percentiles to mitigate the influence of extreme outliers, such as the unprecedented disruptions during the COVID-19 pandemic. Following the literature on macroeconomic predictors (Welch & Goyal, 2008), a 2-month lag structure is used for IP variables, accounting for both the data release schedule and the time required for markets to process new information.

The raw index exhibits substantial growth over the sample period, with the total industrial production index (IP\_TOT) averaging 77.03 (ranging from 39.94 to 104.21), reflecting the long-term expansion of U.S. industrial capacity. After transformation, the series display a mean monthly growth rate of 0.16% with a standard deviation of 0.71%, consistent with typical business cycle fluctuations. The transformed data range from -2.72% to +1.94%, indicating that even after winsorization, the series capture significant variation in production dynamics. The 61 IP series span manufacturing (both durable and non-durable, 23% of the

variables), utilities (5%), mining (2%), and materials/equipment (8%), giving comprehensive coverage of how production dynamics across diverse industrial sectors relate to equity return predictability.

#### 6.1.4 Sales and Inventories: Detailed Construction

Sales and inventory data capture demand signals and supply-demand imbalances at three critical stages of the supply chain: manufacturing (upstream), wholesale trade (midstream), and retail trade (downstream). These three stages provide complementary information: manufacturing shipments and inventories reflect production decisions and intermediate demand; wholesale trade captures distribution channel dynamics; and retail trade reflects final consumer demand. Together, they allow us to trace how demand signals propagate from final consumption back through the supply chain, and how inventory imbalances at each stage signal future production and pricing adjustments. The following subsections detail each data source.

**Manufacturing** The Manufacturing Shipments and Inventories dataset comprises 40 variables measuring two key aspects of manufacturing activity across 20 durable goods industry categories. Sourced from the U.S. Census Bureau’s Manufacturers’ Shipments, Inventories, and Orders (M3) Survey, the data spans March 1992 to July 2025, providing 401 monthly observations. The two data types are: (1) Value of Shipments (MPCVS), representing month-over-month percentage changes in the dollar value of goods shipped, and (2) Total Inventories (MPCTI), representing month-over-month percentage changes in inventory holdings.

Unlike the Industrial Production data, these variables are reported directly as percentage changes and are inherently stationary, requiring no log transformation. The data are winsorized at the 1st and 99th percentiles to mitigate extreme outliers, affecting approximately 1.88% of observations. The 20 industry categories include Primary Metals (31S), Fabricated Metals (32S), Machinery (33S), Computers and Electronics (34S), Transportation Equipment (36S), and various Capital Goods aggregates (TCG, NDE, NXA). Shipments exhibit higher volatility (mean = 0.28%, std = 2.75%) compared to inventories (mean = 0.25%, std = 1.44%), reflecting the more responsive nature of production output relative to inventory management. Following standard practice, a 2-month lag is used for these variables.

Shipments display substantial variation, ranging from -15.1% to +7.2%, capturing both severe contractions (e.g., during recessions) and rapid expansions. Inventories show more moderate fluctuations, ranging from -4.8% to +4.2%, consistent with firms’ tendency to smooth inventory adjustments relative to production changes. This differential volatility

suggests that shipments may provide stronger signals for return predictability, while inventories may capture longer-term strategic positioning by manufacturers.

**Wholesale Trade** The Wholesale Trade Sales and Inventories (WTSI) dataset comprises 42 variables measuring two key aspects of wholesale trade activity across 21 industry categories. Sourced from the U.S. Census Bureau’s Monthly Wholesale Trade Survey, the data spans February 1992 to June 2025, providing 401 monthly observations. The two data types are: (1) Sales (WTSI\_S), representing wholesale sales volumes by industry, and (2) Inventories (WTSI\_I), representing inventory holdings at the wholesale level.

As with the Industrial Production data, the raw dollar values are transformed into log differences:  $\Delta \log(x_t) = \log(x_t) - \log(x_{t-1})$ , which represent month-over-month growth rates and ensure stationarity. The data are winsorized at the 1st and 99th percentiles to mitigate extreme outliers, affecting approximately 2% of observations per variable. The 21 industry categories span both durable goods (e.g., machinery, motor vehicles, furniture) and non-durable goods (e.g., groceries, drugs, petroleum), providing comprehensive coverage of the wholesale distribution channel.

The wholesale trade data show sales exhibiting substantially higher volatility (mean = 0.35%, std = 3.35%) compared to inventories (mean = 0.39%, std = 0.75%), with sales ranging from -8.65% to +10.69% versus inventories ranging from -2.03% to +2.17%. This pattern mirrors the manufacturing data and reflects the more responsive nature of sales relative to inventory management. Wholesale trade variables serve as important leading indicators, as changes in wholesale activity often precede corresponding shifts in retail sales and manufacturing production. Following standard practice, a 2-month lag is used for WTSI variables.

**Retail Trade and Food Services** The Retail and Food Services dataset comprises 54 variables measuring three key aspects of retail trade activity across 36 NAICS-based retail categories. Sourced from the U.S. Census Bureau’s Monthly Retail Trade Survey (MRTS), the data spans March 1992 to June 2025, providing 400 monthly observations. The three data types are: (1) Sales (MPCSM), representing month-over-month percentage changes in retail sales by category (36 variables), (2) Inventories (MPCIM), representing month-over-month percentage changes in retail inventory holdings (9 variables), and (3) Inventory/Sales Ratio (IR), measuring changes in the ratio of inventories to sales, which captures inventory management efficiency (9 variables).

As with the Manufacturing SIO data, these variables are reported directly as percentage changes and are inherently stationary, requiring no log transformation. The data are

winsorized at the 1st and 99th percentiles to mitigate extreme outliers. The 36 retail categories provide comprehensive coverage of the retail sector, including Motor Vehicles and Parts (441), Building Materials (444), Food and Beverage Stores (445), Clothing Stores (448), General Merchandise Stores (452), Department Stores (4522), and numerous other specialized retail segments.

The retail trade data show sales exhibiting moderate volatility (mean = 0.35%, std = 1.07%), ranging from -3.30% to +4.00%, capturing both contractions during economic downturns and expansions during growth periods. Inventories show lower volatility (mean = 0.31%, std = 0.65%), ranging from -2.20% to +1.80%, consistent with retailers' efforts to maintain lean inventory levels and respond quickly to demand shifts. The Inventory/Sales Ratio variables provide additional insight into retailers' strategic positioning and their expectations about future demand. As retail activity is a key component of consumer spending and a coincident indicator of economic conditions, these variables may provide valuable signals for industry return predictability. Following standard practice, a 2-month lag is used for retail variables.

## 6.2 Robustness Analysis

### 6.2.1 Permutation Test of the Network

The procedure in this section is a permutation test in the formal sense: the entire network-construction pipeline (univariate screening at the 5% level across 1,049 signals and 20 industries) is re-run on data in which the null of no predictability is imposed by construction, and the actual, unpermuted result is compared against the resulting empirical distribution. The resimulated draws are referred to as bootstrap draws below, since that is how the underlying resampling is implemented, but the object being tested is the significance pattern of the whole screening procedure, not a single coefficient.

This project examines the predictability of 20 industry returns using 1,049 industry-level time-series signals: the return of each industry is regressed on each of the 1,049 signals, generating a comprehensive set of industry-signal tests.

The first result is that approximately 6.5% of all possible industry return-signal pairs (1,360 of 20,980) are statistically significant, matching Table 9 exactly. Bootstrap simulations show this is above what would be expected under pure chance, though not overwhelmingly so on its own – the sharper evidence comes from the score distribution below.

The second result is that the percentage of significant signals varies across industries, ranging from 5.4% to 8.1%. Bootstrap analysis shows that only the top-performing industries exhibit a share of significant signals that clearly exceeds what would be expected under a

random outcome. For the remaining industries, the results are closer to what noise alone would produce.

The third result concerns the distribution of signal “scores,” defined as the number of industries for which a given signal is statistically significant. The scores range from 0 to 15, meaning that some signals predict up to 15 out of 20 industries. Bootstrap simulations based on 10,000 repetitions show that, under the null hypothesis, no signal predicts more than seven industries at any appreciable probability, and predicting more than four industries occurs only rarely. This evidence suggests that signals identified as systematic are unlikely to be driven by random variation.

**Bootstrap Test of Predictability Under the Null** To assess whether the in-sample predictability patterns could arise by chance, a bootstrap test is implemented that follows the null hypothesis of no forecastability. The predictor panel is held fixed (the full set of predictors) and each return series’ marginal distribution is preserved, but the time ordering of returns is randomized independently for each industry. This destroys any genuine time-series predictive relation between predictors and returns while retaining the unconditional properties of each return series. The same univariate lagged regressions as in the baseline analysis are then re-run:

$$r_{i,t} = \alpha_{ij} + \beta_{ij}x_{j,t-1} + \varepsilon_{ij,t},$$

for each industry  $i$  and predictor  $j$ , and classify significance at the 5% level.

This procedure is repeated 10,000 times to construct empirical null distributions for all outcomes of interest: (i) the percentage of significant predictors by industry, (ii) each predictor’s *score* (number of industries significantly predicted), and (iii) each predictor’s positive and negative scores. Aggregate counts of significant, positive, and negative links are also tracked across the full industry–predictor panel. For each statistic  $T$ , a right-tail empirical  $p$ -value is computed as

$$p = \frac{1}{B} \sum_{b=1}^B \mathbf{1}(T_b^* \geq T^{\text{actual}}),$$

where  $T_b^*$  is the bootstrap draw and  $B = 10000$ .

**Result 1: General** Table 17 shows that the aggregate number of significant coefficients in the actual data (1,360 – the same figure reported in Table 9) is above the bootstrap mean under the null (1,050.75), and the gap is unlikely to arise by chance alone (empirical  $p = 0.0003$ ). The positive-link count (711) is also elevated relative to its null mean (empirical

$p = 0.021$ ), while the negative-link count (649) is only marginally elevated (empirical  $p = 0.076$ ).

Taken together, the global counts already point away from a pure no-forecastability null, though this aggregate test is a blunter instrument than the signal-score test reported next: it cannot say *where* the predictability is concentrated, only that there is more of it than chance would predict. The sharper and more targeted identification evidence – showing that predictive power concentrates in a small set of broadly-propagating signals rather than being spread diffusely across the panel, exactly the opposite of what generic data mining would produce – comes from the industry-level and signal-score results reported next.

**Table 17:** Global Bootstrap Comparison: Actual vs. Null (No Forecastability)

| Statistic               | Actual | Bootstrap Mean | Bootstrap Std. | 5th Pct. | 95th Pct. | Empirical $p$ -value |
|-------------------------|--------|----------------|----------------|----------|-----------|----------------------|
| Total Significant Links | 1,360  | 1050.75        | 89.92          | 919      | 1212      | 0.0003               |
| Total Positive Links    | 711    | 529.59         | 89.03          | 396      | 692       | 0.021                |
| Total Negative Links    | 649    | 521.16         | 89.20          | 390      | 682       | 0.076                |

Notes: Monthly sample uses 20 industry excess returns and 1,049 predictors. Bootstrap uses 10,000 repetitions under the null of no forecastability by independently permuting each industry’s return series over time while keeping predictors fixed. The Actual column corresponds to the full-sample results in Table 9.

**Result 2: Industry Level** Table 18 shows meaningful cross-industry heterogeneity in the share of significant predictors. The highest industries (Arts, Manufacturing, Utilities) have actual significance rates between 6.96% and 7.82%, above their bootstrap means near 5%. However, the corresponding empirical  $p$ -values (0.084 to 0.133) indicate only marginal evidence against the null in these sectors. At the low end, industries such as Professional Services, Other Services, and Admin have actual significance rates around 3.4%–3.8%, below null means near 5%, with large right-tail  $p$ -values (0.75–0.84).

**Table 18:** Industry-Level Percent Significant: Actual vs. Bootstrap Null

| Industry  | Actual (%) | Bootstrap Mean (%) | Bootstrap Std. | 5th Pct. | 95th Pct. | Empirical $p$ -value |
|-----------|------------|--------------------|----------------|----------|-----------|----------------------|
| Arts      | 7.82       | 5.06               | 2.05           | 2.10     | 8.58      | 0.084                |
| Manuf     | 6.96       | 5.04               | 1.89           | 2.19     | 8.39      | 0.133                |
| Utils     | 6.96       | 5.07               | 1.96           | 2.19     | 8.48      | 0.125                |
| Educ      | 6.29       | 4.96               | 1.91           | 2.10     | 8.29      | 0.181                |
| Others    | 6.20       | 4.88               | 1.83           | 2.10     | 8.01      | 0.175                |
| Accom     | 4.19       | 5.00               | 1.93           | 2.10     | 8.39      | 0.618                |
| Health    | 4.00       | 4.99               | 1.82           | 2.00     | 8.20      | 0.697                |
| Admin     | 3.81       | 5.04               | 1.85           | 2.19     | 8.29      | 0.751                |
| Other_ser | 3.53       | 5.04               | 1.96           | 2.00     | 8.58      | 0.836                |
| ProfSer   | 3.43       | 5.02               | 1.82           | 2.00     | 8.10      | 0.840                |

**Result 3: Scores** Under the bootstrap null, predictors are overwhelmingly concentrated at low scores (0–2), with essentially no mass beyond score 6. In contrast, the actual data show a pronounced right tail: multiple predictors with scores from 8 up to 15. These high-score outcomes are virtually absent under the null (empirical  $p$ -values near zero at 10,000 repetitions), indicating that the observed cross-industry concentration of predictive signals is not consistent with random chance. This is the paper’s central piece of evidence that predictability, where it exists, is concentrated in genuinely systematic signals rather than spread diffusely across the predictor panel as data-mining would imply.

One caveat qualifies this evidence at the low end of the distribution. The zero-score bucket is itself larger in the actual data (557) than the null predicts (375.5; empirical  $p < 0.001$ ), and the one- and two-score buckets are correspondingly smaller than the null predicts (empirical  $p$  close to 1). This pattern is consistent with a screening procedure that is mildly conservative for signals with no true predictive content – likely reflecting the cross-sectional correlation among the 20 industry returns discussed in Section 2.2 – while concentrating genuine predictive power sharply among a minority of signals, the opposite of what generic data mining, which would spread spurious hits evenly across the score distribution, would produce. The identifying evidence for genuine diffusion therefore rests specifically on the extreme right tail of the distribution (scores  $\geq 4$ ), not on the low end.

**Table 19:** Predictor Score Distribution: Actual vs. Bootstrap Null

| Score $k$    | Actual # Predictors | Bootstrap Mean | Bootstrap Std. | 5th Pct. | 95th Pct. | Empirical $p$ -value |
|--------------|---------------------|----------------|----------------|----------|-----------|----------------------|
| 0            | 557                 | 375.51         | 35.74          | 316      | 434       | 0.000                |
| 1            | 214                 | 396.01         | 18.10          | 367      | 426       | 0.999                |
| 2            | 118                 | 198.19         | 20.74          | 164      | 234       | 0.999                |
| 3            | 46                  | 62.74          | 12.81          | 43       | 84        | 0.904                |
| 4            | 29                  | 13.89          | 5.21           | 6        | 23        | 0.002                |
| 5            | 24                  | 2.38           | 1.87           | 0        | 6         | 0.000                |
| 6            | 15                  | 0.25           | 0.52           | 0        | 1         | 0.000                |
| 7            | 10                  | 0.01           | 0.10           | 0        | 0         | 0.000                |
| 8            | 8                   | 0.00           | 0.04           | 0        | 0         | 0.000                |
| 9            | 4                   | 0.00           | 0.00           | 0        | 0         | 0.000                |
| 10           | 8                   | 0.00           | 0.00           | 0        | 0         | 0.000                |
| 11           | 6                   | 0.00           | 0.00           | 0        | 0         | 0.000                |
| 12           | 4                   | 0.00           | 0.00           | 0        | 0         | 0.000                |
| 13           | 3                   | 0.00           | 0.00           | 0        | 0         | 0.000                |
| 14           | 2                   | 0.00           | 0.00           | 0        | 0         | 0.000                |
| 15           | 1                   | 0.00           | 0.00           | 0        | 0         | 0.000                |
| <b>Total</b> | <b>1,049</b>        | <b>1,049.0</b> |                |          |           |                      |

Notes: A predictor’s score is the number of industries for which it is significant at 5% in univariate lagged regressions. Bootstrap null is based on 10,000 repetitions with independent permutation of each industry return series over time (predictors fixed). The Actual column spans all 1,049 signals; see Section 3.1.1 and Figure 1 for the corresponding zero-score and non-zero-score breakdown.

### 6.2.2 Additional Identification Checks

The permutation test in Section 6.2.1 addresses whether the network reflects genuine cross-sectional structure rather than chance. It does not, by itself, address whether that structure is stable over time, tied to a particular crisis episode, or an artifact of how returns are weighted or industries are classified. Table 20 outlines ten further robustness dimensions, each following the same logic applied to a different aspect of the data (sample period, weighting, classification, or lag). The recursive and rolling out-of-sample design is the most direct extension of the OOS evidence in Section 3.3 and is detailed below.

**Table 20:** Additional Robustness Dimensions for Future Extension

| Check                                           | What it does                                                                                    | What it rules out                                                               |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Recursive (expanding-window) OOS                | Re-estimates at each month from Feb. 2015–Dec. 2023, one-step-ahead forecast each time (§6.2.2) | Accuracy concentrated in one subperiod of the holdout                           |
| Rolling (120-month window) OOS                  | Same as above but trailing window only, discarding old data                                     | Distinguishes a time-varying network from a stable one                          |
| Random-network placebo                          | Randomly reassigns 1,049 signals to sectors 1,000+ times                                        | Predictability driven by signal informativeness alone, not true I-O linkages    |
| Recession/expansion subsamples                  | Re-estimates separately on NBER recession-only vs. expansion-only months                        | Demand/cost sign pattern being a business-cycle artifact                        |
| Crisis exclusion                                | Drops Great Recession and COVID windows entirely, re-estimates on what remains                  | Results driven by one or two extreme episodes                                   |
| Pre-/post-2000 subsamples                       | Splits sample at Jan. 2000, re-estimates each half                                              | Channel signs being specific to one era                                         |
| Equal- vs. value-weighted returns               | Re-builds sector returns as equal-weighted portfolios                                           | Results being a large-firm-only phenomenon                                      |
| Fama–French portfolios                          | Re-runs screening/network construction on FF industry groupings                                 | Results being specific to the NAICS classification                              |
| Alternative lags (0, 3, 6 months)               | Re-estimates network at each lag                                                                | Traces the diffusion half-life; rules out look-ahead bias at lag 0              |
| PLS component sensitivity ( $K = 1\text{--}5$ ) | Re-estimates the headline specification varying components, each fixed ex ante                  | Whether $K = 2$ specifically, versus small $K$ generally, is driving the result |

*Note:* International extension to non-U.S. production networks (e.g., OECD Inter-Country Input-Output tables) is not included above since it requires a different data infrastructure entirely and is left as future work rather than a near-term robustness check.

**Recursive and rolling out-of-sample forecasts.** The single 70/30 split used in Section 3.3 evaluates the network at one point in time. A recursive design instead re-estimates

the full pipeline at each point  $t_0 \in \{\text{Feb. 2015}, \dots, \text{Dec. 2023}\}$  using only data through  $t_0$ , forecasts one month ahead, and repeats – yielding roughly 106 one-step-ahead forecast errors per sector rather than one aggregate OOS  $R^2$ , and testing directly whether accuracy is concentrated in a subperiod. A rolling design repeats this using only a trailing 120-month window at each step: if rolling outperforms expanding, the network is time-varying and old data actively hurts; if the two are similar, the network is stable and accuracy differences come from elsewhere.

### 6.2.3 Cross-Sector Summary: Idiosyncratic Signal Overfitting

Table ?? in Section 3.3 already shows that idiosyncratic signals contribute meaningfully to out-of-sample fit for several sectors (e.g., Wholesale, Construction, Arts). Table ?? below reports, for all 19 non-market sectors, the idiosyncratic-signal in-sample  $R^2$ , out-of-sample  $R^2$ , and the gap between them, drawn from the idiosyncratic-signal columns of Table ??. Three sectors show *negative* out-of-sample  $R^2$  despite substantial in-sample fit – Education (−16.3%), Professional Services (−15.6%), and Finance (−2.4%) – meaning their idiosyncratic signal sets are, at best, uninformative out of sample and at worst worse than a constant-mean forecast. A further five sectors show overfitting gaps above 20 percentage points beyond those three – Health (33.3 pts), Utilities (24.4 pts), Agriculture (22.5 pts), Mining (21.6 pts), and Other Services (20.7 pts) – while Professional Services and Education post the two largest gaps of all (42.0 and 49.7 points, respectively). This table is reported precisely because it complicates the paper’s headline result; the honest reading is that idiosyncratic predictability is real but concentrated in a minority of sectors, not present uniformly across the panel.

□□□

**Table 21:** Out-of-Sample  $R^2$  Performance on Common Stable Signals (%)

| Sector        | Ridge       | Elastic Net | LASSO       | PLS 2       | PLS 3        | PLS 5        |
|---------------|-------------|-------------|-------------|-------------|--------------|--------------|
| Arts          | 13.90       | 11.39       | 9.33        | 11.81       | 5.14         | 2.04         |
| Finance       | 12.78       | 8.46        | 6.63        | 3.75        | 4.73         | 1.70         |
| Const         | 12.43       | 11.77       | 10.84       | 12.77       | 10.83        | 10.31        |
| Utils         | 11.87       | 11.53       | 11.07       | 1.04        | 7.52         | 6.39         |
| Educ          | 11.19       | 3.44        | -1.87       | -0.18       | -3.50        | -9.36        |
| Retail        | 10.76       | 8.85        | 6.71        | 7.29        | 7.86         | 3.57         |
| Info          | 10.01       | 5.10        | 2.14        | 1.81        | 0.33         | -1.47        |
| Market        | 8.81        | 6.84        | 5.78        | 4.38        | 5.05         | 3.07         |
| Others        | 7.87        | 4.85        | 2.90        | 5.44        | 3.00         | -1.99        |
| Manuf         | 7.71        | 6.01        | 5.04        | 4.09        | 3.65         | 0.59         |
| Mining        | 7.57        | 4.53        | 3.12        | 4.21        | 1.87         | 0.02         |
| Wholes        | 7.35        | 3.08        | 0.52        | 2.86        | -4.75        | -5.01        |
| Trans         | 7.34        | 5.53        | 4.91        | 5.69        | 3.76         | 2.98         |
| Admin         | 7.14        | 4.96        | 3.96        | 0.66        | 1.15         | 0.27         |
| Other Ser     | 6.51        | 1.52        | -0.72       | -2.22       | -3.55        | -6.87        |
| Accom         | 5.72        | 3.59        | 1.15        | -1.46       | -2.57        | -3.45        |
| Realest       | 3.14        | -0.40       | -2.49       | -1.20       | -3.84        | -4.53        |
| Agric         | 0.41        | -12.04      | -19.14      | -15.75      | -16.88       | -28.51       |
| Profser       | -2.86       | -10.14      | -12.82      | -18.38      | -20.99       | -22.96       |
| Health        | -3.43       | -7.53       | -11.08      | -18.92      | -19.78       | -22.46       |
| <b>Mean</b>   | <b>7.31</b> | <b>3.57</b> | <b>1.30</b> | <b>0.39</b> | <b>-1.05</b> | <b>-3.78</b> |
| <b>Median</b> | <b>7.64</b> | <b>4.91</b> | <b>3.01</b> | <b>2.33</b> | <b>1.51</b>  | <b>-0.73</b> |

Common Stable Signals = predictors significant in both training (1992-2014) and full sample (1992-2024).

70/30 train-test split. All methods use standardized predictors. Ridge uses all stable signals with L2 regularization.

LASSO and Elastic Net perform variable selection. PLS uses 2, 3, or 5 components for dimensionality reduction.

Negative values indicate worse performance than a naive historical mean forecast.

## References

- Acemoglu, D., Carvalho, V. M., Ozdaglar, A., & Tahbaz-Salehi, A. (2012). The network origins of aggregate fluctuations. *Econometrica*, 80(5), 1977–2016.
- Bessembinder, H., Burt, A., & Hrdlicka, C. M. (2025). Can we tame the factor zoo? the roles of alternative databases and extraction methods. *Can we tame the factor zoo*.
- Blanchard, O. J. (1986). The wage price spiral. *The Quarterly Journal of Economics*, 101(3), 543–565.
- Blinder, A. S. (1982). Inventories and the structure of macro models. *The American Economic Review*, 72(2), 11–16.
- Breeden, D. T. (1979). An intertemporal asset pricing model with stochastic consumption and investment opportunities. *Journal of Financial Economics*, 7(3), 265–296.
- Cohen, L., & Frazzini, A. (2008). Economic links and predictable returns. *The Journal of Finance*, 63(4), 1977–2011.

- Fama, E. F., & French, K. R. (1997). Industry costs of equity. *Journal of financial economics*, 43(2), 153–193.
- Fama, E. F., & French, K. R. (2012). Size, value, and momentum in international stock returns. *Journal of Financial Economics*, 105(3), 457–472.
- Gofman, M., Segal, G., & Wu, Y. (2020). Dynamic input-output networks. *Review of Financial Studies*, 33(11), 5193–5237.
- Hamermesh, D. S. (1993). *Labor demand*. Princeton University Press.
- Harvey, C. R., Liu, Y., & Zhu, H. (2016). ...and the cross-section of expected returns. *The Review of Financial Studies*, 29(1), 5–68.
- Herskovic, B., Kelly, B., Lustig, H., & Van Nieuwerburgh, S. (2020). The common factor in idiosyncratic volatility: Quantitative asset pricing implications. *Journal of Financial Economics*, 136(3), 714–750.
- Hoberg, G., & Phillips, G. (2016). Text-based network industries and endogenous product differentiation. *Journal of political economy*, 124(5), 1423–1465.
- Hong, H., Torous, W., & Valkanov, R. (2007). Do industries lead stock markets? *Journal of Financial Economics*, 83(2), 367–396.
- Hou, K., Xue, C., & Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3), 650–705.
- Kelly, B., & Pruitt, S. (2013). Market expectations in the cross-section of present values. *The Journal of Finance*, 68(5), 1721–1756.
- Keynes, J. M. (1936). *The general theory of employment, interest and money*. Palgrave Macmillan.
- Khan, A., & Thomas, J. K. (2007). Inventories and the business cycle: An equilibrium analysis of (s, s) policies. *American Economic Review*, 97(4), 1165–1188.
- Leontief, W. W. (1936). Quantitative input and output relations in the economic systems of the united states. *The Review of Economics and Statistics*, 18(3), 105–125.
- Lucas, R. E. (1978). Asset prices in an exchange economy. *Econometrica*, 46(6), 1429–1445.
- Menzly, L., & Ozbas, O. (2010). Market segmentation and cross-predictability of returns. *The Journal of Finance*, 65(4), 1555–1580.
- Savov, A. (2011). Asset pricing with garbage. *The Journal of Finance*, 66(1), 177–201.
- Welch, I., & Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *The Review of Financial Studies*, 21(4), 1455–1508.